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(54) **OPTIMIZED CLOSED-LOOP CONTROL OF
A COMBUSTION APPARATUS**

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(71) Applicant: **Siemens Aktiengesellschaft, München**
(DE)

(72) Inventor: **Alexander Meier, Rastatt (DE)**

(57)

ABSTRACT

(73) Assignee: **Siemens Aktiengesellschaft, München**
(DE)

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Various embodiments of the teachings herein include a method for control of a combustion apparatus. An example includes loading a first characteristic curve from a memory; determining a current value of a performance variable; determining a first input value of the variable for an open-loop control mode; assigning the first input value to a first speed and/or to a first position using the first characteristic curve; determining a first open-loop control signal as a function of the first speed and/or the first position; and sending the first open-loop control signal to the first actuator, wherein the first open-loop control signal causes the first actuator to change a combustion variable including the air supply and/or the fuel supply.

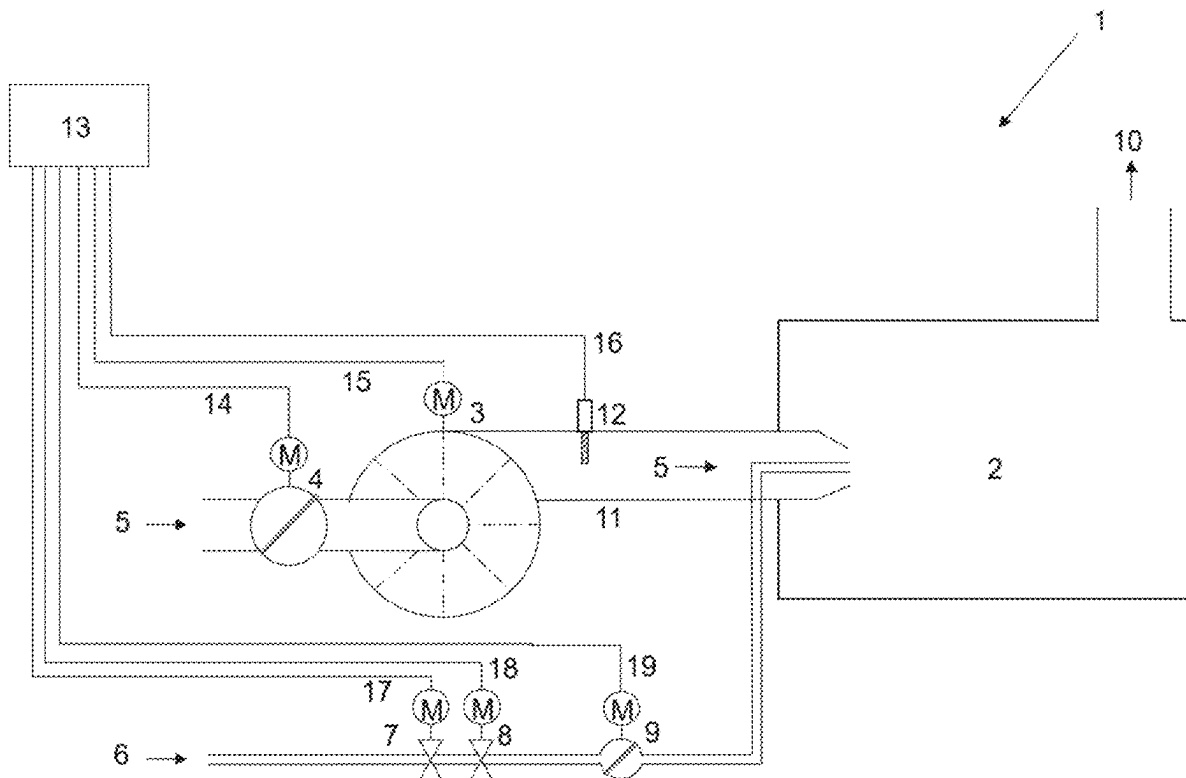


FIG 1

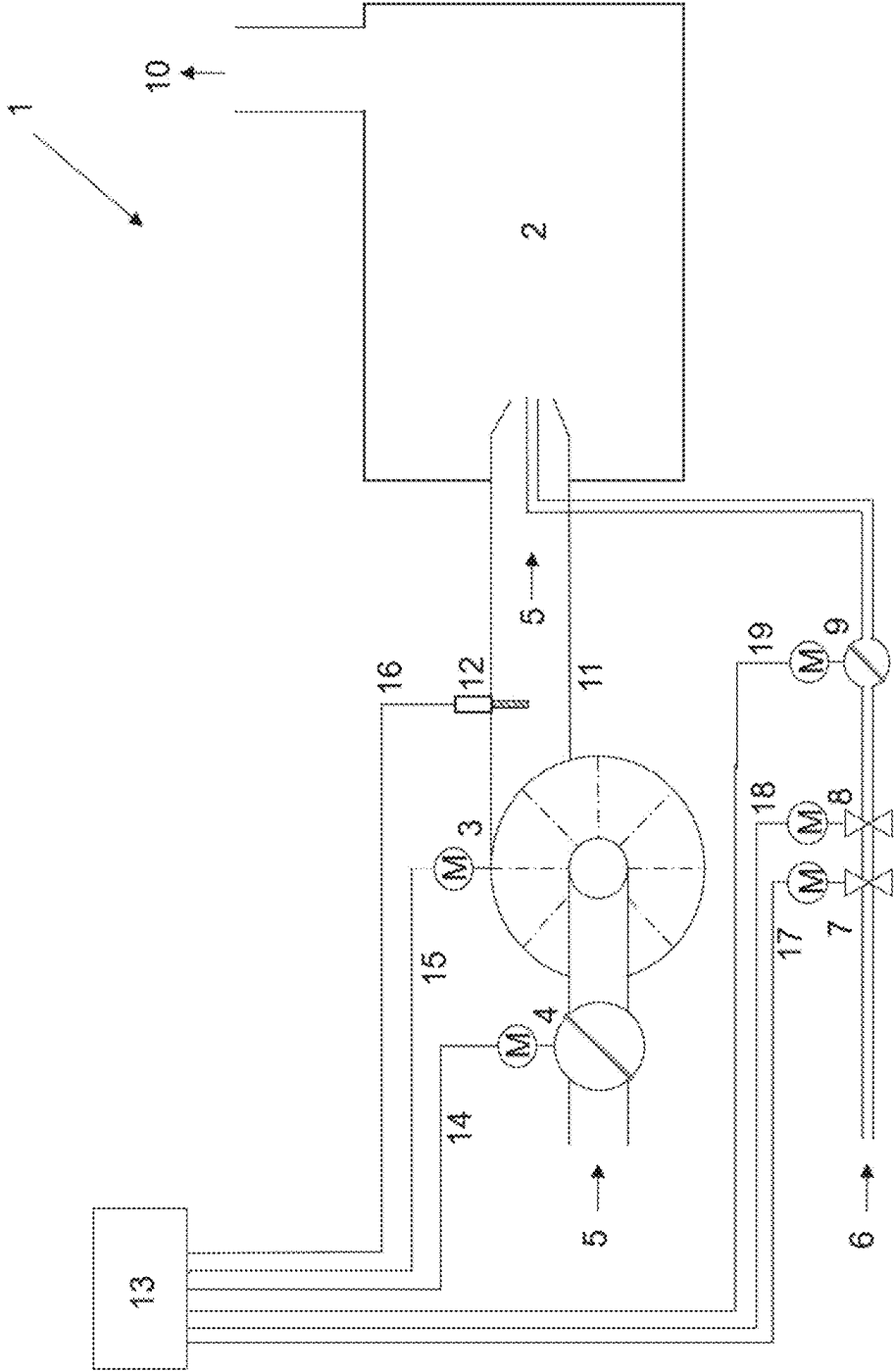


FIG 2

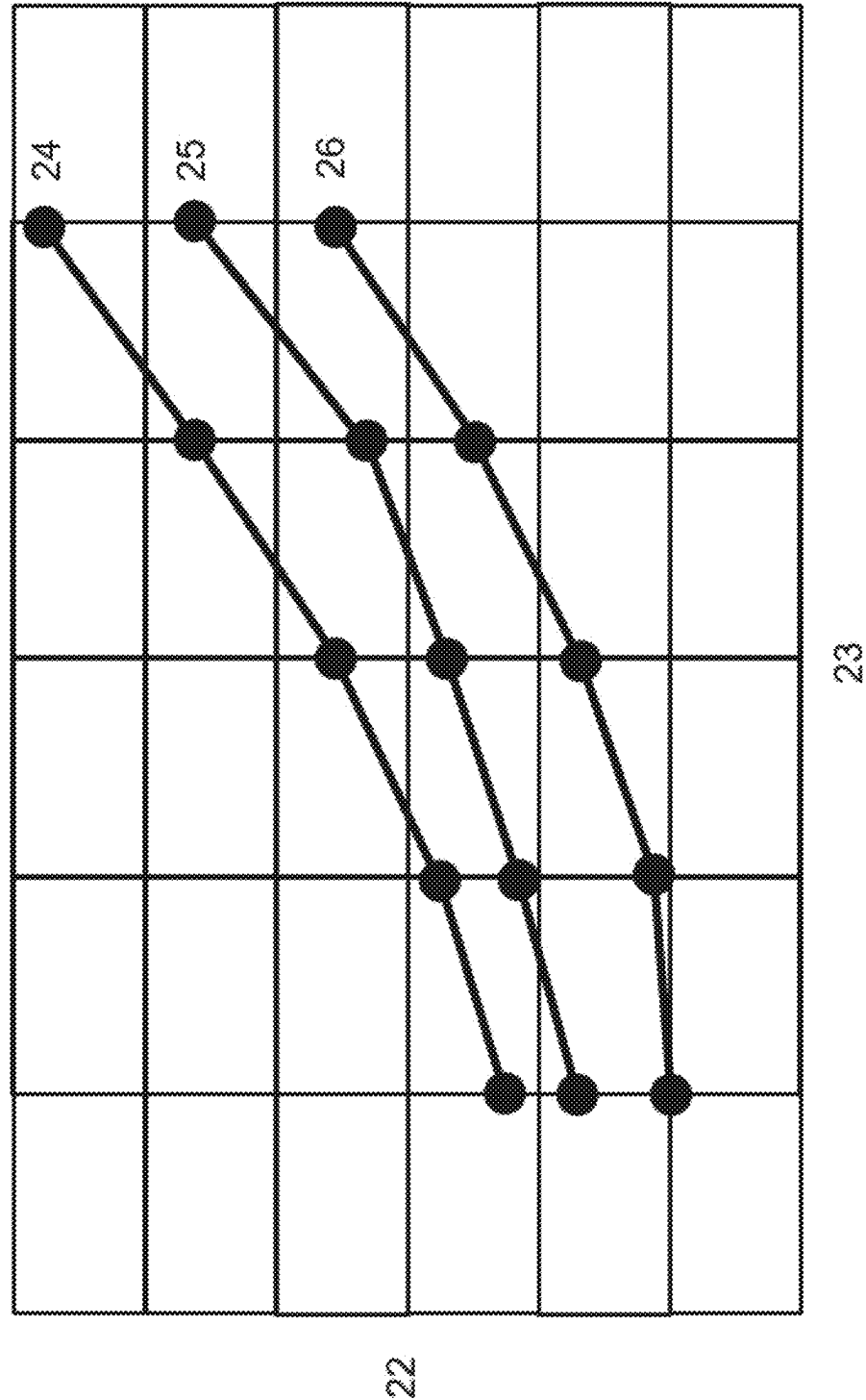


FIG 4

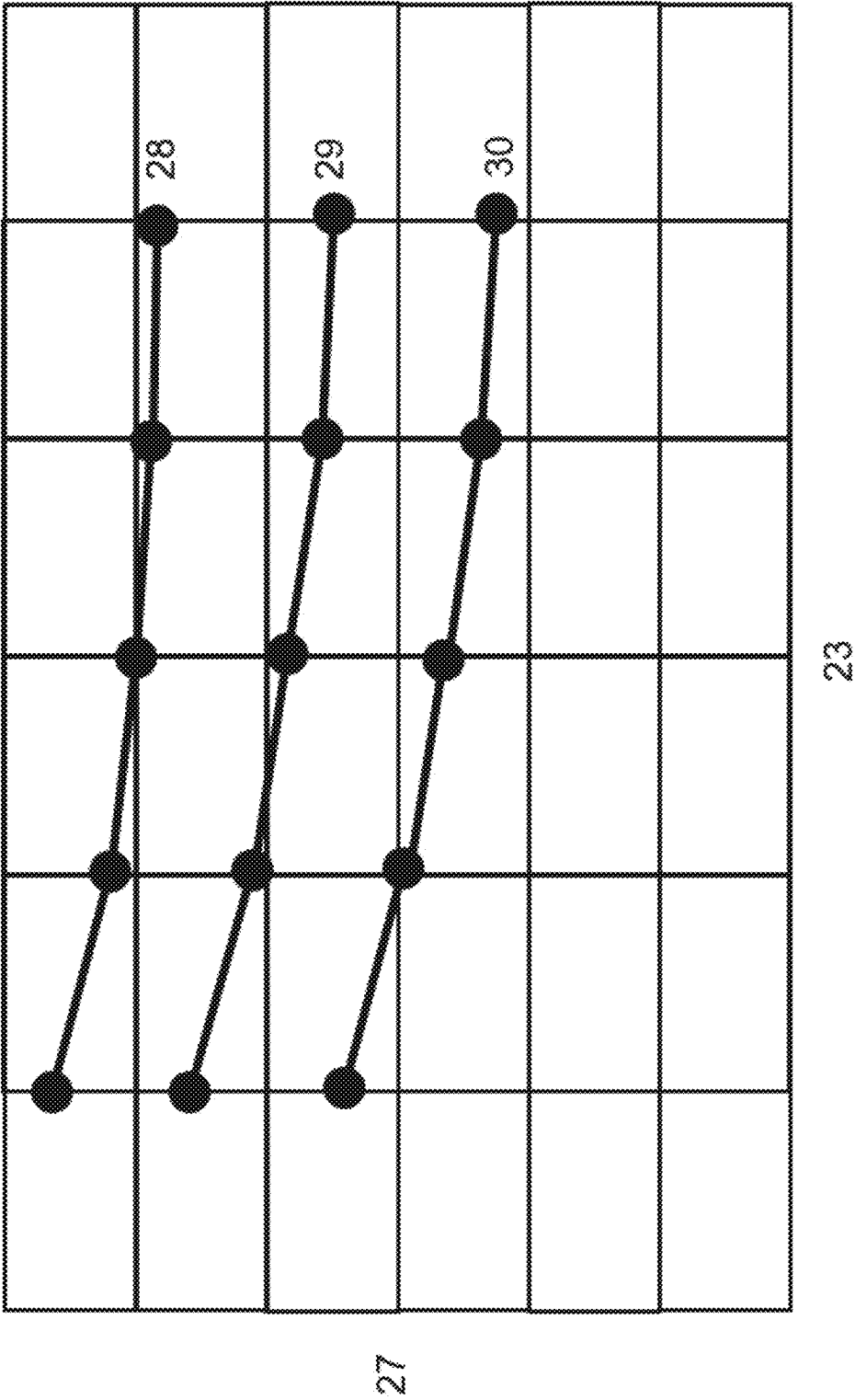


FIG 5

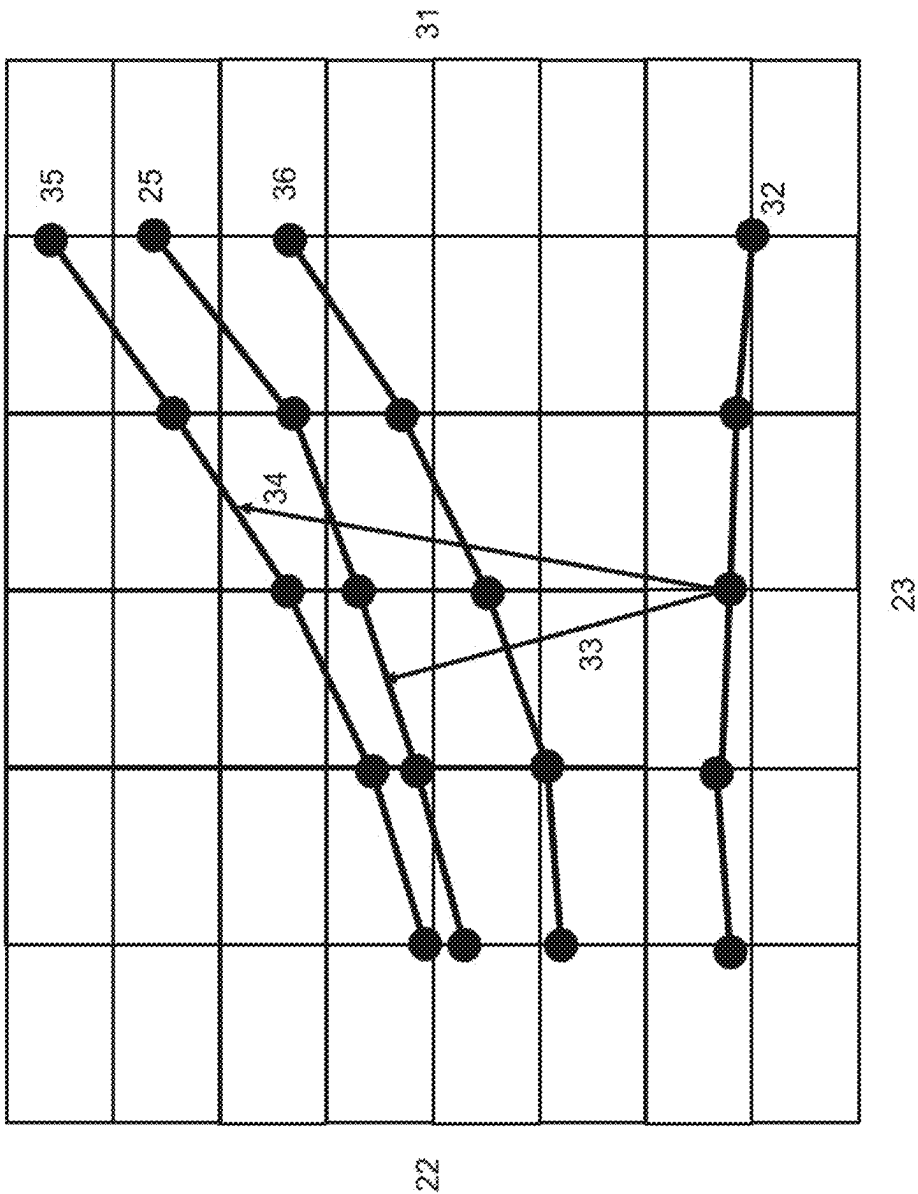
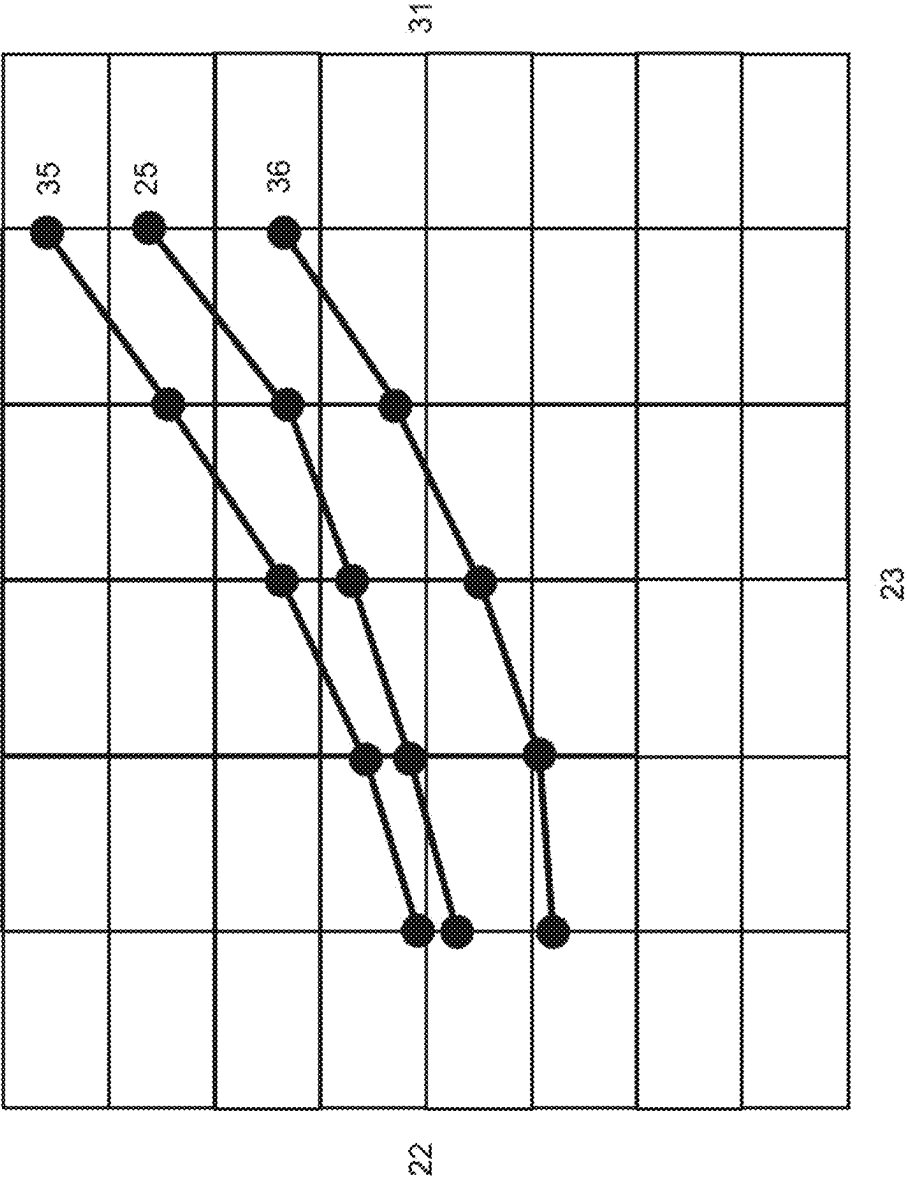


FIG 6



OPTIMIZED CLOSED-LOOP CONTROL OF A COMBUSTION APPARATUS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to EP Application No. 24162652.2 filed Mar. 11, 2024, the contents of which are hereby incorporated by reference in their entirety.

TECHNICAL FIELD

[0002] The present disclosure is related to closed-loop and/or open-loop control of a combustion apparatus. Various embodiments of the teachings herein include systems and/or methods for automation of a combustion apparatus as a function of an oxygen concentration.

BACKGROUND

[0003] The ratio of fuel to air has to be adjusted during operation of a combustion apparatus. In some prior art systems, the air actuator characteristic curve(s) and fuel characteristic curve(s) are ascertained via the performance during the adjusting procedure. For example, the characteristic curves can be ascertained from a low performance to a maximum performance, or also vice versa. The air ratio λ is adjusted for each performance point in the process. Air supply sensors can also be used in support. Popular air supply sensors are based on speed, mass flow, differential pressure, air-volume flow, etc. The absolute performance is then determined via a measurement of the fuel supply at one point at least or at a plurality of points. The burner performance is assigned to the respective characteristic curve points with the aid of the heat value H_u of the fuel currently being fed in. The performance values of the other characteristic curve points are determined by interpolation, preferably by linear interpolation. Determination of the performance values of the other characteristic curve points by measuring can be considered, moreover.

[0004] In some other systems, the air actuator characteristic curves and the fuel characteristic curves are predefined. In most cases the characteristic curves are empirically ascertained in the laboratory. The burner performance is predefined by a fixed function from one of the two characteristic curves. Different characteristic curves and/or sets of characteristic curves, which are likewise predefined, are used for different fuels.

[0005] In some other systems, the change in a fuel composition or change in air density is revealed by means of a λ -sensor. This can be, for example, an O_2 probe in the waste gas from which λ is calculated directly. For example, an ionization electrode, whose signal is evaluated accordingly, can also be used. To keep the air ratio λ constant, either the air supply can be changed or else the fuel supply can be corrected until the λ -sensor measures the original value of an air ratio λ again. If the at least one air supply signal is readjusted to keep the air ratio λ constant, then the burner performance at this characteristic curve point nearly always changes with the fuel composition too. If the fuel supply signal is readjusted to keep the air ratio λ constant, the burner performance changes dependent on the fuel. In order to adjust the performance, a new characteristic curve of the air actuator has to be manually or automatically selected or calculated for the case of a performance correction.

[0006] Common types of gas in combustion facilities are those from the E-gas group (to EN 437:2009-09) and gases from the B/P-gas group (to EN 437:2009-09). Gases from the E-gas group, like nearly all gases from the second gas family (to EN 437:2009-09), contain methane as the main component. Gases from the B/P-gas group, like all gases from the third gas family (to EN 437:2009-09), have propane gas as their basis. The mixtures based on methane gas or propane gas ultimately represent mixtures from different gas sources with which the combustion apparatus can be supplied.

[0007] As a rule, characteristic curves are provided for different types of gas, which curves are selected in situ during commissioning in accordance with the available gas group. The adjustment is made, for example, by selecting one or more curve(s) stored in the memory of a closed-loop control unit. The adjustment can also be made using a set of parameters stored in the memory of the closed-loop control unit or using a plurality of sets of parameters stored in the memory of the closed-loop control unit. Those characteristic curves reproduce the characteristic of the quantity of fuel supplied to the combustion chamber in relation to the supplied quantity of air. Instead of the quantity of supplied air, the speed of a fan in the air supply of the combustion apparatus can be plotted. Further, the position and/or the control signal of an air damper can be considered as a measure of the air supply. In addition, the air supply can be ascertained using a mass flow sensor which can be arranged, for example, in a side duct. An apparatus comprising a mass flow sensor in a side duct is disclosed, for example, in the European patent application EP3301363A1.

[0008] The characteristic curves can be stored, for example, in table form with linear interpolation or also with the aid of polynomials as a mathematical function. This form of the characteristic curve assignment is disclosed in the European patent EP3299718B1 which was granted on Oct. 30, 2019. An application EP3299718A1 relating to European patent EP3299718B1 was filed on Sep. 21, 2016. The European patent EP3299718B1 does not claim priority.

[0009] A quantity of air is suitable as a performance value if air temperature, air pressure or air humidity change only insubstantially or are acquired metrologically. The effects of air temperature and air pressure are taken into account when the quantity of air is measured with an air flow sensor. The effect of the air humidity is of secondary importance primarily at lower temperatures.

[0010] VAILLANT GmbH patent application EP2682679A2 is concerned with a method for closed-loop control and/or monitoring of a fuel gas-powered burner. EP2682679A2 describes the start-up of working points below and above a desired air ratio. A signal of a mass flow sensor, which is arranged in a duct between an air line and a fuel gas line, is subsequently recorded. A correct or incorrect adjustment of the system is inferred from the signal.

[0011] GAZ DE FRANCE, FR European patent application EP0326494A1 describing an apparatus for measuring the heat capacity of a flow of fuel. A heat value of a fuel is ascertained on the basis of the signals of a mass flow sensor and an ionization sensor.

[0012] Karl Dungs GmbH & Co. KG patent application DE102013106987A1 describes a method and an apparatus for determining a fuel value variable and a gas-powered facility with an apparatus of this kind. For this, there is a fuel

value sensor present in a combustion chamber of the apparatus, and this comprises an ionization sensor and preferably a temperature sensor.

[0013] Gaswärme-Institut e.V. in Essen patent application DE102006051883A1 describes a facility and a method for the adjustment, open-loop control or closed-loop control of the fuel/combustion air ratio for operating a burner. A heat value or a Wobbe index is automatically ascertained during the course of adjustment, open-loop control or closed-loop control.

[0014] EON RUHRGAS AG a European patent application EP1467149A1 describes a method for monitoring combustion in a combustion facility. A probe, such as an oxygen probe, can be used in a waste gas duct of the combustion facility. Should a combustion air ratio lie outside a pre-defined limit, the combustion apparatus is switched off.

[0015] SIEMENS AG European patent application EP4050258A1 describes ascertaining performance of a combustion apparatus using a fuel parameter. A fuel parameter is provided during the course of ascertaining the performance. The combustion apparatus is subject to closed-loop control accordingly.

SUMMARY

[0016] The teachings of the present disclosure include flexible automation of a combustion apparatus without and with taking into account feedback from a sensor. In particular, the manner and the further development of the combustion inside the combustion apparatus during closed-loop control and/or open-loop control of the apparatus are to be taken into account. For example, some embodiments of the teachings herein include a method for control of a combustion apparatus (1), the combustion apparatus (1) comprising a combustion chamber (2), an air supply duct (11) leading to the combustion chamber (2), a fuel supply duct leading to the combustion chamber (2), at least one first actuator (3, 4, 7-9) selected from at least one air actuator (3, 4), which acts on an air supply V_{L} through the air supply duct (11), and at least one fuel actuator (7-9), which acts on a fuel supply V_{B} through the fuel supply duct, the combustion apparatus (1) comprising a waste gas path (10), at least one oxygen-based sensor (20) in the waste gas path (10) and a closed-loop control facility and/or open-loop control facility and/or monitoring facility (13) with a memory in which at least one first characteristic curve (25, 35, 36) which indicates for the at least one first actuator (3, 4, 7-9) a first speed characteristic and/or a first position characteristic with respect to a performance-based variable (23), and a change, which is different from zero, are stored, the method comprising: loading the at least one first characteristic curve (25, 35, 36) and the change from the memory; determining a first, current value of the performance-based variable (23); determining a first input value of the performance-based variable (23) for the open-loop control mode as a function of the first, current value of the performance-based variable (23) and as a function of the change, wherein the determination occurs independently of an oxygen-based signal of the at least one sensor (20); assigning the first input value of the performance-based variable (23) for the open-loop control mode to a first speed and/or to a first position using the at least one first characteristic curve (25, 35, 36); determining a first open-loop control signal as a function of the first speed and/or the first position; and sending the first open-loop

control signal to the at least one first actuator (3, 4, 7-9), wherein the first open-loop control signal causes the at least one first actuator (3, 4, 7-9) to change at least one combustion variable selected from the air supply V_{L} and/or the fuel supply V_{B} .

[0017] In some embodiments, the method further comprises: starting combustion in the combustion apparatus (1); and sending the first open-loop control signal to the at least one first actuator (3, 4, 7-9) within five seconds of the start of combustion.

[0018] In some embodiments, the change is different from zero and is constant, and the method further comprises determining the first input value of the performance-based variable (23) for the open-loop control mode as a function of the first, current value of the performance-based variable (23) and as a function of the first, constant change, wherein the determination occurs independently of the oxygen-based signal of the at least one sensor (20).

[0019] In some embodiments, at least one change characteristic curve (32), which indicates a change characteristic with respect to the performance-based variable (23), is stored in the memory, and the method further comprises: loading the at least one change characteristic curve (32) from the memory; determining the change from the first, current value of the performance-based variable (23) by assigning the first, current value of the performance-based variable (23) using the at least one change characteristic curve (32); and determining the first input value of the performance-based variable (23) for the open-loop control mode as a function of the first, current value of the performance-based variable (23) and as a function of the change from the first, current value of the performance-based variable (23), wherein the determination occurs independently of the oxygen-based signal of the at least one sensor (20).

[0020] In some embodiments, the method further comprises determining the first input value of the performance-based variable (23) for the open-loop control mode as the sole function of the first, current value of the performance-based variable (23) and the change.

[0021] In some embodiments, the method further comprises determining the first input value of the performance-based variable (23) for the open-loop control mode as the sum of the first, current value of the performance-based variable (23) and the change, wherein the determination occurs independently of the oxygen-based signal of the at least one sensor (20).

[0022] In some embodiments, at least one desired value characteristic curve (29), which indicates a characteristic of a residual oxygen content and/or an oxygen concentration and/or an oxygen partial pressure with respect to the performance-based variable (23), is stored in the memory, and the method further comprises: after the first open-loop control signal has been sent, recording at least one oxygen-based signal, which indicates the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure, by way of the at least one sensor (20); sending the oxygen-based signal to the closed-loop control facility and/or open-loop control facility and/or monitoring facility (13); determining a measured value by way of the closed-loop control facility and/or open-loop control facility and/or monitoring facility (13) on the basis of the at least one oxygen-based signal; determining a second, current value of the performance-based variable (23); loading the at least one desired value characteristic curve (29) from the memory;

assigning the second, current value of the performance-based variable (23) to a desired value (27) using the at least one desired value characteristic curve (29); comparing the measured value with the desired value (27); generating a closed-loop control signal on the basis of the comparison between the measured value and the desired value (27); and sending the closed-loop control signal to the at least one first actuator (3, 4, 7-9), wherein the closed-loop control signal causes the at least one first actuator (3, 4, 7-9) to change the at least one combustion variable.

[0023] In some embodiments, the combustion apparatus (1) comprises at least one second actuator (7-9, 3, 4) selected from the at least one fuel actuator (7-9) and the at least one air actuator (3, 4), wherein the at least one second actuator (7-9, 3, 4) is different from the at least one first actuator (3, 4, 7-9), wherein at least one second characteristic curve (35, 36, 25), which for the at least one second actuator (7-9, 3, 4) indicates a second speed characteristic and/or a second position characteristic with respect to the performance-based variable (23), is stored in the memory, and the method further comprises: assigning the second, current value of the performance-based variable (23) to a second speed and/or to a second position using the at least one second characteristic curve (35, 36, 25), wherein the assignment takes place independently of the change; determining a second open-loop control signal as a function of the second speed and/or the second position; and sending the second open-loop control signal to the at least one second actuator (7-9, 3, 4), wherein the second open-loop control signal causes the at least one second actuator (7-9, 3, 4) to change the at least one combustion variable.

[0024] In some embodiments, the method further comprises determining a second, current value of the performance-based variable (23) from the first, current value of the performance-based variable (23).

[0025] In some embodiments, the method further comprises determining a second, current value of the performance-based variable (23) after the first open-loop control signal has been sent.

[0026] In some embodiments, the method further comprises: checking the measured value for at least one error; if checking of the measured value yields the at least one error: determining a third, current value of the performance-based variable (23); determining a further input value of the performance-based variable (23) for the open-loop control mode as a function of the second or third current value of the performance-based variable (23) and as a function of the change, wherein the determination occurs independently of the oxygen-based signal of the at least one sensor (20); assigning the further input value of the performance-based variable (23) for the open-loop control mode to a third speed and/or to a third position using the at least one first characteristic curve (25, 35, 36); determining an emergency open-loop control signal as a function of the third speed and/or the third position; and sending the emergency open-loop control signal to the at least one first actuator (3, 4, 7-9), wherein the emergency open-loop control signal causes the at least one first actuator (3, 4, 7-9) to change the at least one combustion variable.

[0027] In some embodiments, a limit value for the measured value is stored in the memory, and the method further comprises: checking the measured value for the at least one error by comparing the measured value with the limit value;

and identifying the at least one error if the measured value is smaller than the limit value.

[0028] As another example, some embodiments of the teachings herein include a combustion apparatus (1) comprising a combustion chamber (2), an air supply duct (11) leading to the combustion chamber (2), a fuel supply duct leading to the combustion chamber (2), at least one first actuator (3, 4, 7-9) selected from at least one air actuator (3, 4), which acts on an air supply V_{L} through the air supply duct (11), and at least one fuel actuator (7-9), which acts on a fuel supply V_{B} through the fuel supply duct, the combustion apparatus (1) comprising at least one second actuator (7-9, 3, 4) selected from the at least one fuel actuator (7-9) and the at least one air actuator (3, 4), wherein the at least one second actuator (7-9, 3, 4) is different from the at least one first actuator (3, 4, 7-9), the combustion apparatus (1) comprising a waste gas path (10), at least one oxygen-based sensor (20) in the waste gas path (10) and a closed-loop control facility and/or open-loop control facility and/or monitoring facility (13) with a memory in which at least one first characteristic curve (25, 35, 36), which indicates for the at least one first actuator (3, 4, 7-9) a first speed characteristic and/or a first position characteristic with respect to a performance-based variable (23), and at least one second characteristic curve (35, 36, 25), which indicates for the at least one second actuator (7-9, 3, 4) a second speed characteristic and/or a second position characteristic with respect to the performance-based variable (23), and a change, which is different from zero, and a limit value for a measured value are stored, wherein the closed-loop control facility and/or open-loop control facility and/or monitoring facility (13) is communicatively connected to the at least one first actuator (3, 4, 7-9) and to the at least one second actuator (7-9, 3, 4) and to the at least one oxygen-based sensor (20) and to the memory and is configured to carry out one or more of the methods described herein.

[0029] As another example, some embodiments include a computer program comprising instructions which cause the closed-loop control facility and/or open-loop control facility and/or monitoring facility (13) of the combustion apparatus (1) as described herein to execute one or more of the methods described herein.

[0030] As another example, some embodiments include a computer-readable medium on which a computer program as described herein is stored.

BRIEF DESCRIPTION OF THE DRAWINGS

[0031] Various details are accessible to a person skilled in the art on the basis of the following detailed description. The individual embodiments are not limiting to the scope of the disclosure. The drawings, which accompany the description, may be described as follows:

[0032] FIG. 1 schematically shows a combustion apparatus incorporating teachings of the present disclosure;

[0033] FIG. 2 illustrates closed-loop and/or open-loop control of a combustion apparatus without inclusion of a signal corresponding to an oxygen concentration and/or corresponding to an oxygen partial pressure incorporating teachings of the present disclosure;

[0034] FIG. 3 shows a combustion apparatus with O_2 sensor for detection in the waste gas incorporating teachings of the present disclosure;

[0035] FIG. 4 shows a characteristic of an oxygen concentration and/or an oxygen partial pressure with respect to a performance-based variable for combustion incorporating teachings of the present disclosure;

[0036] FIG. 5 illustrates closed-loop and/or open-loop control of a combustion apparatus with the inclusion of a signal corresponding to an oxygen concentration and/or corresponding to an oxygen partial pressure incorporating teachings of the present disclosure; and

[0037] FIG. 6 illustrates a further closed-loop and/or open-loop control of a combustion apparatus with the inclusion of a signal corresponding to an oxygen concentration and/or corresponding to an oxygen partial pressure.

DETAILED DESCRIPTION

[0038] The present disclosure describes automation of a combustion apparatus using an oxygen concentration in a waste gas path of the combustion apparatus. For this, an oxygen-based sensor is arranged in or on the waste gas path of the combustion apparatus such that it is exposed to a waste gas flow. The automation of the combustion apparatus can comprise open-loop control and/or closed-loop control of combustion in the combustion apparatus. A transition from an initial open-loop control mode into a subsequent closed-loop control mode occurs in the process.

[0039] The combustion apparatus comprises a closed-loop control facility and/or open-loop control facility and/or monitoring facility. The closed-loop control facility and/or open-loop control facility and/or monitoring facility automates the combustion in the combustion apparatus using at least one actuator. The actuator can be an air actuator or a fuel actuator and influences the supply of air or fuel to a combustion chamber of the combustion apparatus.

[0040] A first characteristic curve is primarily stored in the closed-loop control facility and/or open-loop control facility and/or monitoring facility for a closed-loop control mode of the combustion apparatus. The first characteristic curve relates to said actuator of the combustion apparatus. It indicates a speed and/or a position of the actuator with respect to a performance-based variable.

[0041] The first characteristic curve for a closed-loop control mode is also used in the open-loop control mode. For this, in open-loop control mode a working point is displaced along the characteristic curve such that combustion avoiding harmful emissions is ensured. This thus enables the combustion apparatus to be started safely.

[0042] After some time a closed-loop control mode follows the open-loop control mode. In closed-loop control mode the combustion apparatus is subject to closed-loop control using said sensor. That sensor provides a signal which is used as a feedback signal with closed-loop control by way of the actuator. A mixed closed-loop and open-loop control mode is possible in this case. That is to say, a first actuator is subject to closed-loop control. A second actuator is subject to open-loop control. Preferably, the working point displacement like at the beginning of combustion does not occur with open-loop control of the second actuator.

[0043] Finally, the closed-loop control characteristic curve permits an emergency mode. Such an emergency mode can occur, for example, if the sensor fails. Closed-loop control is consequently no longer possible. Safer operation of the combustion apparatus is then enabled using the closed-loop control characteristic curve and with displacement of the working point.

[0044] FIG. 1 shows a combustion apparatus 1, such as a wall-hung gas burner and/or an oil burner with a combustion chamber 2 incorporating teachings of the present disclosure. The heat generator exchanges the heat energy of the hot fuels and/or combustion gases into a different fluid, such as water. For example a hot water heating system is operated and/or drinking water is heated using the warm water. In some embodiments, an item, for example in an industrial process, can be heated using the heat energy of the hot combustion gases. Further, the heat generator can serve to heat water in a plant for obtaining lithium and/or lithium carbonate. The waste gases are removed from the combustion chamber 2 for example via a waste gas stack and/or a flue gas stack and/or a flue 10.

[0045] The supply air 5 for the combustion process is supplied to the combustion chamber 2 of the combustion apparatus 1 via a (motor-) driven fan 3. Via the signal line 15 the closed-loop control facility and/or open-loop control facility and/or monitoring facility 13 specifies to the fan 3 the air supply $\dot{V}_L$ that it should convey. The fan speed thus becomes a measure of the conveyed quantity of air and/or of the air supply $\dot{V}_L$.

[0046] In some embodiments, the fan 3 reports the fan speed back to the closed-loop control facility and/or open-loop control facility and/or monitoring facility 13. The speed of the fan 3 may consequently be mapped multiple times to a supplied quantity of air. If the quantity of air is adjusted via an air damper 4 and/or a valve, the damper and/or valve position and/or the measured value derived from the signal of a mass flow sensor 12 and/or volume flow sensor can be used as a measure of the quantity of air. The sensor may be arranged in the duct for the air supply $\dot{V}_L$. In some embodiments, the sensor provides a signal which is converted into a flow measured value using a suitable signal processing unit. A signal processing facility ideally comprises at least one analog-to-digital converter. In some embodiments, the signal processing facility, in particular the analog-to-digital converter(s), is integrated in the closed-loop control facility and/or open-loop control facility and/or monitoring facility 13.

[0047] In some embodiments, the closed-loop control facility and/or open-loop control facility and/or monitoring facility 13 comprises a delta-sigma converter. The delta-sigma converter enables a conversion of analog signals, for example of the mass flow sensor 12, into digital values. The delta-sigma converter can be an integral component of the closed-loop control facility and/or open-loop control facility and/or monitoring facility 13.

[0048] With regard to compactness, the delta-sigma converter and the closed-loop control facility and/or open-loop control facility and/or monitoring facility 13 can be parts of the same system-on-a-chip.

[0049] The measured value of a pressure sensor and/or a mass flow sensor 12 in a side duct can also be used as a measure of the air supply $\dot{V}_L$. A combustion apparatus with supply duct and side duct is disclosed, for example, in the European patent EP3301364B1. A combustion facility with supply duct and side duct is claimed, wherein a mass flow sensor projects into the supply duct.

[0050] The sensor 12 ascertains a signal which corresponds to the pressure value dependent on the air supply $\dot{V}_L$ and/or the air flow (particle and/or mass flow) in the side duct. The sensor 12 provides a signal which is converted into a measured value using a suitable signal processing facility.

In some embodiments, signals of a plurality of sensors are converted into a shared measured value. A suitable signal processing facility ideally comprises at least one analog-to-digital converter. In some embodiments, the signal processing facility, in particular the analog-to-digital converter(s), is integrated in the closed-loop control facility and/or open-loop control facility and/or monitoring facility 13.

[0051] In some embodiments, the air supply $\dot{V}_L$ is the value of the current air throughput rate. The air throughput rate can be measured and/or indicated in cubic meters of air per hour. The air supply $\dot{V}_L$ can be measured and/or indicated in cubic meters of air per hour. The air supply $\dot{V}_L$ can also be measured and/or indicated in cubic feet of air per minute.

[0052] Mass flow sensors 12 allow measurement in the case of large flow velocities specifically in connection with combustion apparatuses 1 during operation. Typical values of such flow velocities lie in the ranges between 0.1 meters per second and five meters per second, ten meters per second, fifteen meters per second, twenty meters per second, or even one hundred meters per second. Mass flow sensors, which are suitable for the present disclosure, are, for example, OMRON® D6F-W or SENSOR TECHNICS® WBA type sensors. The usable range of these sensors typically begins at velocities between 0.01 meters per second and 0.1 meters per second and ends at a velocity such as five meters per second, ten meters per second, fifteen meters per second, twenty meters per second, or even one hundred meters per second. In other words, lower limits, such as 0.1 meters per second, can be combined with upper limits, such as five meters per second, ten meters per second, fifteen meters per second, twenty meters per second, or even one hundred meters per second.

[0053] The fuel supply $\dot{V}_B$ is adjusted and/or corrected by the closed-loop control facility and/or open-loop control facility and/or monitoring facility 13 with the aid of a fuel actuator and/or a (motor-) adjustable valve. In the design in FIG. 1 the fuel comprises a combustion gas. A combustion apparatus 1 can then be connected to different combustion gas sources, for example to sources with a high methane content and/or to sources with a high propane content. In FIG. 1 the quantity of combustion gas through a fuel actuator 7-9 is adjusted by the closed-loop control facility and/or open-loop control facility and/or monitoring facility 13. The fuel actuator 7-9 can comprise or be, for example, a (motor-) adjustable fuel valve 9. The actuation value 19, for example with a pulse width-modulated signal, of the gas valve is a measure of the quantity of combustion gas. In a further embodiment actuation is via a CAN bus. The fuel can also comprise an oil or an oil mixture.

[0054] The actuation value is also a value for the fuel supply $\dot{V}_B$. In some embodiments, the fuel valve 9 is adjusted using a stepping motor. In any case the step setting of the stepping motor is a measure of the quantity of combustion gas. The fuel valve 9 can also be integrated in a unit with at least one or two of the safety shut-off valves 7 or 8. Furthermore, the fuel valve 9 can be a valve subject to internal closed-loop control via a flow sensor, which valve receives a desired value 19 and adjusts the actual value of the through-flow sensor to the desired value. The through-flow sensor can be implemented as a volume flow sensor for example as a turbine wheel counter, bellows-type gas flowmeter and/or as a differential pressure sensor. The

through-flow sensor can also be designed as a mass flow sensor, for example as a thermal mass flow sensor.

[0055] If a gas damper is used as the actuator 9, then the position of the damper can be used as a measure of the quantity of combustion gas. In some embodiments, a measured value derived from the signal of a mass flow sensor and/or a volume flow sensor can be used as a measure of the quantity of combustion gas. That sensor is advantageously arranged in the supply duct for fuel. That sensor generates a signal, which is converted into a flow measured value (measured value of the particle and/or mass flow and/or volume flow) using a suitable signal processing facility.

[0056] A suitable signal processing facility ideally comprises at least one analog-to-digital converter. In some embodiments, the signal processing facility, in particular the analog-to-digital converter(s), is integrated in the closed-loop control facility and/or open-loop control facility and/or monitoring facility 13.

[0057] In some embodiments, the closed-loop control facility and/or open-loop control facility and/or monitoring facility 13 comprises a delta-sigma converter. The delta-sigma converter enables a conversion of analog signals, for example of the mass flow sensor or volume flow sensor for combustion gas, into digital values. The delta-sigma converter can be an integral component of the closed-loop control facility and/or open-loop control facility and/or monitoring facility 13. With regard to compactness, the delta-sigma converter and the closed-loop control facility and/or open-loop control facility and/or monitoring facility 13 can be parts of the same system-on-a-chip.

[0058] A person skilled in the art recognizes that the above-mentioned values can also be calculated from a combination of variables ascertained by sensors. Those values are then measures of the supply (particle and/or mass flow and/or volume flow) of combustion gas. A person skilled in the art also recognizes that the fuel supply of a liquid fuel can be ascertained in a similar manner.

[0059] The combustion apparatus 1 in FIG. 1 may be operated without closed-loop control for an oxygen-based variable such as the O_2 concentration in the waste gas. In this regard settings and/or speeds 22 of different actuators are plotted against the burner performance 23 in FIG. 2. In some embodiments, the burner performance 23 is a current performance of the combustion apparatus 1. In some embodiments, the burner performance 23 is a current heating performance of the combustion apparatus.

[0060] Specifically, FIG. 2 shows a characteristic curve 24 for the motor-driven fan 3. Characteristic curve 24 consequently illustrates a speed of the motor-driven fan 3 against the burner performance 23. Characteristic curve 24 illustrates a speed of the motor-driven fan 3 against a current burner performance 23.

[0061] FIG. 2 shows a characteristic curve 25 for a fuel actuator 7-9. Characteristic curve 25 consequently illustrates a position of the fuel actuator 7-9 against the burner performance 23. In some embodiments, characteristic curve 25 illustrates a position of the fuel actuator 7-9 against a current burner performance 23. The fuel actuator 7-9 can comprise or be a fuel damper. In this case characteristic curve 25 illustrates a position of the fuel damper against the burner performance 23. In particular, characteristic curve 25 in this case illustrates a position of the fuel damper against a current burner performance 23.

[0062] FIG. 2 also shows a characteristic curve 26 for an air damper 4. The characteristic curve 26 consequently indicates a position of the air damper 4 against the burner performance 23. In some embodiments, characteristic curve 26 illustrates a position of the air damper 4 against a current burner performance 23.

[0063] In FIG. 2 each of the actuators 3, 4, 7-9 are subject to open-loop control in terms of their speeds and/or positions for a predefined burner performance 23. In particular, the actuators 3, 4, 7-9 in FIG. 2 can be subject to open-loop control in terms of their speeds and/or positions for a current burner performance 23. Open-loop control is such that an adequate excess of air exists for each burner performance 23. That is to say, the air ratio λ is always greater than one, where $\lambda=1$ corresponds to a stoichiometric combustion. Harmful emissions are avoided in this way.

[0064] One drawback of open-loop control according to FIG. 2 is that the excess of air is not adjusted. This means there is no reduction in the fan speed and no change in the position of the air damper 4 so that combustion is optimized after the start. In particular, an efficiency of the combustion apparatus 1 is not optimized by a reduction in the fan speed or by a change in the position of the air damper 4.

[0065] Similarly, the actuators 3, 4, 7-9 in FIG. 2 can each be subject to closed-loop control in terms of their speeds and/or positions. In particular, the actuators 3, 4, 7-9 in FIG. 2 can be subject to closed-loop control in terms of their speeds and/or positions for a current burner performance 23. Closed-loop control is such that an adequate excess of air exists for each burner performance 23. This means the air ratio λ is always greater than one, where $\lambda=1$ corresponds to a stoichiometric combustion. Harmful emissions are avoided in this way.

[0066] One drawback of closed-loop control according to FIG. 2 is that the excess of air is not adjusted. This means there is no reduction in the fan speed and no change in the position of the air damper 4 so combustion is optimized after the start. In particular, an efficiency of the combustion apparatus 1 is not optimized by a reduction in the fan speed or by a change in the position of the air damper 4.

[0067] FIG. 3 shows a combustion apparatus 1 with a sensor 20 for detecting an oxygen concentration and/or an oxygen partial pressure. In particular, the sensor 20 can detect or indicate a residual oxygen content. The sensor 20 comprises, for example, an O_2 sensor. In some embodiments, the sensor 20 is an O_2 sensor. The sensor 20 can be arranged, for example, in a waste gas stack and/or a flue gas stack and/or a flue 10.

[0068] In some embodiments, the sensor 20 is a gas sensor for recording a signal which indicates an oxygen concentration and/or an oxygen partial pressure. The sensor 20 can also be embodied for recording a signal which corresponds to at least one further gas. The further gas comprises, for example, an oxidizable gas in a waste gas stack and/or a flue gas stack and/or a flue 10 of the combustion apparatus 1.

[0069] In some embodiments, the sensor 20 can also be embodied for recording a signal which corresponds to a concentration of carbon monoxide. In some embodiments, the sensor 20 can be embodied for recording a signal which corresponds to a temperature.

[0070] In some embodiments, the sensor 20 comprises at least one disk, which comprises at least ninety percent by weight zirconium oxide. Further, the disk comprises at least one further element or a further compound selected from:

[0071] yttrium oxide and

[0072] hafnium oxide.

[0073] In addition, the exemplary sensor 20 comprises at least three electrodes, including two electrodes made from doped platinum. The doped platinum has between half a percent and fifteen percent by weight zirconium oxide. One electrode of the three electrodes comprises a gold alloy. One of the electrodes made from doped platinum and the electrode comprising a gold alloy are arranged on a first side of the disk. A further one of the electrodes made from doped platinum is arranged on a second side of the disk, with the second side being different from the first side. The second side of the disk opposes the first side of the disk. The second side of the disk adjoins a sealed chamber of the sensor 20.

[0074] The sensor 20 for recording a signal, which indicates a partial pressure of oxygen and/or a residual oxygen content and/or an oxygen concentration, generates a signal 21. The signal 21 is read by the closed-loop control facility and/or open-loop control facility and/or monitoring facility 13 and appropriately processed. With the aid of the signal 21 it is possible to correct to a desired value for each fan speed and/or for each air supply $\dot{V}_L$ and/or for each burner performance. The desired value relates to a partial pressure of oxygen and/or a residual oxygen content and/or an oxygen concentration. In particular, the desired value can relate to a partial pressure of oxygen and/or a residual oxygen content and/or an oxygen concentration in a waste gas stack. Further, the desired value can relate to a partial pressure of oxygen and/or a residual oxygen content and/or an oxygen concentration in a flue gas stack. Furthermore, the desired value can relate to a partial pressure of oxygen and/or a residual oxygen content and/or an oxygen concentration in a flue 10.

[0075] In some embodiments, the suitable signal processing facility for acquiring and evaluating the signal 21 of the sensor 20 comprises at least one analog-to-digital converter. In some embodiments, the signal processing facility, in particular the analog-to-digital converter(s), is integrated in the closed-loop control facility and/or open-loop control facility and/or monitoring facility 13.

[0076] In some embodiments, the closed-loop control facility and/or open-loop control facility and/or monitoring facility 13 comprises a delta-sigma converter. The delta-sigma converter enables a conversion of analog signals, for example of the sensor 20, into digital values. The delta-sigma converter can be an integral component of the closed-loop control facility and/or open-loop control facility and/or monitoring facility 13. With regard to compactness, the delta-sigma converter and the closed-loop control facility and/or open-loop control facility and/or monitoring facility 13 can be parts of the same system-on-a-chip.

[0077] FIG. 4 illustrates a characteristic of a residual oxygen content and/or an oxygen concentration and/or an oxygen partial pressure with respect to a burner performance 23. In particular, FIG. 4 illustrates a characteristic of a residual oxygen content and/or an oxygen concentration and/or an oxygen partial pressure with respect to a current burner performance 23.

[0078] Values 27 of the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure between a minimum value and a maximum value are plotted in FIG. 4. The minimum value of the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure marks the bottom end of the vertical axis. The maximum value of the residual oxygen content and/or

the oxygen concentration and/or the oxygen partial pressure marks the top end of the vertical axis.

[0079] The minimum value of the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure can lie, for example, at zero percent. The maximum value of the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure can lie, for example, between five and ten percent. In particular, the maximum value of the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure can lie between six and nine percent. Further, the maximum value of the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure can lie between six and eight percent, for example at six percent.

[0080] Plotted on the horizontal axis in FIG. 4 is a performance-related variable selected from

[0081] the burner performance,

[0082] the current burner performance.

[0083] The burner performance 23 or the current burner performance 23 preferably relate to the combustion apparatus 1. The burner performance 23 is a heating performance of the combustion apparatus 1. In some embodiments, the current burner performance 23 is a current heating performance of the combustion apparatus 1.

[0084] The values 23 from a minimum value of the performance-based variable to a maximum value of the performance-based variable are plotted. The values 23 of the performance-based variable increase from left to right. The minimum value of the performance-based variable marks the left end of the horizontal axis. The maximum value of the performance-based variable marks the right end of the horizontal axis.

[0085] The minimum value of the performance-based variable along the horizontal axis in FIG. 4 can lie, for example, at zero percent of a nominal performance of the combustion apparatus 1. The nominal performance of the combustion apparatus 1 relates to a nominal heating performance of the combustion apparatus 1. The maximum value of the performance-based variable along the horizontal axis can lie, for example, between one hundred and one hundred and fifty percent of the nominal performance of the combustion apparatus 1. In some embodiments, the maximum value of the performance-based variable along the horizontal axis can lie, for example, between one hundred and ten and one hundred and thirty percent of the nominal performance of the combustion apparatus 1. Further, the maximum value of the performance-based variable along the horizontal axis can lie, for example, between one hundred and fifteen and one hundred and twenty-five percent of the nominal performance of the combustion apparatus 1, for example at one hundred and twenty percent.

[0086] FIG. 4 shows three characteristic curves and/or curves 28-30 of the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure for combustion apparatuses 1. A first characteristic curve and/or a first characteristic 28 indicates maximum residual oxygen contents and/or maximum oxygen concentrations and/or maximum oxygen partial pressures. The oxygen concentrations of the first characteristic 28 in FIG. 4 are indicated as a percentage of total molecules. The partial pressures of the first characteristic 28 are indicated in FIG. 4 as a percentage of a total pressure. The same applies to the vertical axis.

[0087] A minimum value of the performance-based variable forms part of the first characteristic curve and/or the

first characteristic 28. All other values of the performance-based variable along the first characteristic curve and/or along the first characteristic 28 are greater than that minimum value of the performance-based variable.

[0088] The minimum value of the performance-based variable of the first characteristic curve and/or the first characteristic 28 is generally not identical to the corresponding minimum value along the horizontal axis. The minimum value of the performance-based variable of the first characteristic curve and/or the first characteristic 28 can lie, for example, between ten and thirty percent of the nominal performance of the combustion apparatus 1. In particular, the minimum value of the performance-based variable of the first characteristic curve and/or of the first characteristic 28 can lie at twenty percent of the nominal performance of the combustion apparatus 1. In the example in FIG. 4 the minimum value of the performance-based variable of the first characteristic curve and/or of the first characteristic 28 lies at twenty percent of the nominal performance.

[0089] A maximum value of the performance-based variable forms part of the first characteristic curve and/or the first characteristic 28. All other values of the performance-based variable along the first characteristic curve and/or along the first characteristic 28 are lower and/or smaller than that maximum value of the performance-based variable.

[0090] The maximum value of the performance-based variable of the first characteristic curve and/or of the first characteristic 28 is generally not identical to the corresponding maximum value along the horizontal axis. The maximum value of the performance-based variable of the first characteristic curve and/or of the first characteristic 28 can lie, for example, between ninety and one hundred and ten percent of the nominal performance of the combustion apparatus 1. The nominal performance of the combustion apparatus 1 relates to a nominal heating performance of the combustion apparatus 1. In particular, the maximum value of the performance-based variable of the first characteristic curve and/or of the first characteristic 28 can lie at one hundred percent of the nominal performance of the combustion apparatus 1. In the example in FIG. 4 the maximum value of the performance-based variable of the first characteristic curve and/or of the first characteristic 28 lies at one hundred percent of the nominal performance.

[0091] (Three) further points are plotted along the first characteristic curve and/or along the first characteristic 28 between the minimum value of the performance-based variable and the corresponding maximum value. The further points are points and/or values 27 of the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure with respect to the performance-based variable 23. Similarly, that point which forms part of the minimum value of the performance-based variable, is a point and/or value 27 of the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure with respect to the performance-based variable 23. Further, that point which forms part of the maximum value of the performance-based variable is a point and/or value 27 of the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure with respect to the performance-based variable 23.

[0092] It is possible to interpolate between the points of the first characteristic curve and/or of the first characteristic 28. For example, it is possible to linearly interpolate between the points of the first characteristic curve and/or the

first characteristic **28**. Further, maximum values of the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure can be determined using cubic splines for values **23** of the performance-based variable. In particular, maximum values of the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure can be calculated using cubic splines for values **23** of the performance-based variable.

[0093] The first characteristic curve and/or the first characteristic **28** can be a mathematical relationship, such as a polynomial, moreover. Maximum values of the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure are thus calculated using the mathematical relationship for values **23** of the performance-based variable. In particular, maximum values of the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure are calculated using a polynomial for values **23** of the performance-based variable. The polynomial corresponds to the first characteristic curve and/or the first characteristic **28**.

[0094] The first characteristic curve and/or the first characteristic **28** reproduces a first maximum characteristic curve and/or a first maximum characteristic of the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure. That is to say, the closed-loop control facility and/or open-loop control facility and/or monitoring facility **13** is or can be communicatively connected to a sensor **20**, for example to an oxygen sensor **20**, of the combustion apparatus **1**. In addition, the closed-loop control facility and/or open-loop control facility and/or monitoring facility **13** is or can be communicatively connected to at least one fuel actuator **7-9** of the combustion apparatus **1**. A first characteristic curve and/or a first characteristic **28** is stored in a memory, for example in a non-volatile memory, of the closed-loop control facility and/or open-loop control facility and/or monitoring facility **13**. The closed-loop control facility and/or open-loop control facility and/or monitoring facility **13** is embodied:

[0095] to receive one or more signal(s), which indicate (s) a residual oxygen content and/or an oxygen concentration and/or an oxygen partial pressure, from the sensor **20**;

[0096] to determine a measured value of the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure on the basis of the one or more signal(s), which indicate(s) the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure;

[0097] to load the first characteristic curve and/or the first characteristic **28** from the memory;

[0098] using the first characteristic curve and/or using the first characteristic **28**, to map a burner performance **23**, in particular a current burner performance **23**, to a comparative value **27** of the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure;

[0099] to compare the comparative value **27** with the measured value; and

[0100] if the measured value is greater than the comparative value **27**;

[0101] to generate a closure signal and send it to the at least one fuel actuator **7-9**, wherein the closure signal causes the at least one fuel actuator **7-9** to close when the at least one fuel actuator **7-9** receives the signal.

[0102] A second characteristic curve and/or a second characteristic **29** indicates desired values of the residual oxygen content and/or the oxygen concentration and/or of the oxygen partial pressure. The oxygen concentrations of the second characteristic **29** are indicated in FIG. 4 as a percentage of total molecules. The partial pressures of the second characteristic **29** are indicated in FIG. 4 as a percentage of a total pressure.

[0103] A minimum value of the performance-based variable forms part of the second characteristic curve and/or the second characteristic **29**. All other values of the performance-based variable along the second characteristic curve and/or along the second characteristic **29** are greater than that minimum value of the performance-based variable.

[0104] The minimum value of the performance-based variable of the second characteristic curve and/or the second characteristic **29** is generally not identical to the corresponding minimum value along the horizontal axis. The minimum value of the performance-based variable of the second characteristic curve and/or the second characteristic **29** can lie, for example, between ten and thirty percent of the nominal performance of the combustion apparatus **1**. In particular, the minimum value of the performance-based variable of the second characteristic curve and/or the second characteristic **29** can lie at twenty percent of the nominal performance of the combustion apparatus **1**. Preferably, the minimum value of the performance-based variable of the second characteristic curve and/or the second characteristic **29** is identical to the corresponding minimum value of the first characteristic curve and/or of the first characteristic **28**. In the example in FIG. 4 the minimum value of the performance-based variable of the second characteristic curve and/or the second characteristic **29** lies at twenty percent of the nominal performance.

[0105] A maximum value of the performance-based variable forms part of the second characteristic curve and/or the second characteristic **29**. All other values of the performance-based variable along the second characteristic curve and/or along the second characteristic **29** are lower and/or smaller than that maximum value of the performance-based variable.

[0106] The maximum value of the performance-based variable of the second characteristic curve and/or the second characteristic **29** is generally not identical to the corresponding maximum value along the horizontal axis. The maximum value of the performance-based variable of the second characteristic curve and/or the second characteristic **29** can lie, for example, between ninety and one hundred and ten percent of the nominal performance of the combustion apparatus **1**. In particular, the maximum value of the performance-based variable of the second characteristic curve and/or of the second characteristic **29** can lie at one hundred percent of the nominal performance of the combustion apparatus **1**. Preferably, the maximum value of the performance-based variable of the second characteristic curve and/or the second characteristic **29** is identical to the corresponding maximum value of the first characteristic curve and/or the first characteristic **28**. In the example in FIG. 4 the maximum value of the performance-based variable of the second characteristic curve and/or the second characteristic **29** lies at one hundred percent of the nominal performance.

[0107] (Three) further points are plotted along the second characteristic curve and/or the second characteristic **29** between the minimum value of the performance-based vari-

able and the corresponding maximum value. The further points are points and/or values 27 of the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure with respect to the performance-based variable 23. Similarly, that point which forms part of the minimum value of the performance-based variable, is a point and/or value 27 of the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure with respect to the performance-based variable 23. Further, that point which forms part of the maximum value of the performance-based variable, is a point and/or value 27 of the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure with respect to the performance-based variable 23.

[0108] It is possible to interpolate between the points of the second characteristic curve and/or the second characteristic 29. For example, it is possible to linearly interpolate between the points of the second characteristic curve and/or the second characteristic 29. Further, desired values 27 of the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure can be determined using cubic splines for values 23 of the performance-based variable. In particular, desired values 27 of the residual oxygen content and/or the oxygen concentration and/or of the oxygen partial pressure can be calculated using cubic splines for values 23 of the performance-based variable.

[0109] The second characteristic curve and/or the second characteristic 29 can be a mathematical relationship, such as a polynomial. Desired values 27 of the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure are thus calculated using the mathematical relationship for values 23 of the performance-based variable. In particular, desired values 27 of the residual oxygen content and/or the oxygen concentration and/or of the oxygen partial pressure are calculated using a polynomial for values 23 of the performance-based variable. The polynomial corresponds to the second characteristic curve and/or the second characteristic 29.

[0110] The second characteristic curve and/or the second characteristic 29 reproduces a desired value characteristic of the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure.

[0111] That is to say, the closed-loop control facility and/or open-loop control facility and/or monitoring facility 13 is or can be communicatively connected to a sensor 20, for example to an oxygen sensor 20, of the combustion apparatus 1. In addition, the closed-loop control facility and/or open-loop control facility and/or monitoring facility 13 is or can be communicatively connected to at least one actuator 3, 4, 7-9 of the combustion apparatus 1. The at least one actuator 3, 4, 7-9 of the combustion apparatus 1 is selected from:

- [0112] a fuel actuator 7-9 of the combustion apparatus 1, wherein the fuel actuator 7-9 acts on a fuel supply to the combustion apparatus 1,
 - [0113] a fan 3 of the combustion apparatus 1, wherein the fan 3 acts on an air supply to the combustion apparatus 1,
 - [0114] an air damper 4 of the combustion apparatus 1, wherein the air damper 4 acts on an air supply to the combustion apparatus 1.
- [0115] The preceding list of actuators 3, 4, 7-9 is not intended to be exhaustive.

[0116] A second characteristic curve and/or a second characteristic 29 is stored in a memory, for example in a non-volatile memory, of the closed-loop control facility and/or open-loop control facility and/or monitoring facility 13. In some embodiments, the closed-loop control facility and/or open-loop control facility and/or monitoring facility 13 is embodied:

- [0117] to receive one or more signal(s), which indicate(s) a residual oxygen content and/or an oxygen concentration and/or an oxygen partial pressure, from the sensor 20;
 - [0118] to determine a measured value of the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure on the basis of the one or more signal(s), which indicate(s) the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure;
 - [0119] to load the second characteristic curve and/or the second characteristic 29 from the memory;
 - [0120] to map a burner performance 23, in particular a current burner performance 23, to a desired value 27 of the residual oxygen content and/or the oxygen concentration and/or of the oxygen partial pressure using the second characteristic curve and/or using the second characteristic 29;
 - [0121] to compare the desired value 27 with the measured value; and
 - [0122] to generate a closed-loop control signal on the basis of the comparison between desired value 27 and measured value and to send the closed-loop control signal to the at least one actuator 3, 4, 7-9.
- [0123] In some embodiments, the closed-loop control signal causes the at least one actuator 3, 4, 7-9 to change a combustion variable selected from:
- [0124] a fuel supply to the combustion apparatus 1 if the at least one actuator is or comprises a fuel actuator 7-9,
 - [0125] an air supply to the combustion apparatus 1 if the at least one actuator is or comprises a fan 3 or an air damper 4.

[0126] The combustion variable is ideally changed in such a way that subsequent measured values, which are determined from subsequently received signals, which indicate the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure, approximate the desired value 27.

[0127] A third characteristic curve and/or a third characteristic 30 indicates minimum residual oxygen contents and/or minimum oxygen concentrations and/or minimum oxygen partial pressures. The oxygen concentrations of the third characteristic 30 are indicated in FIG. 4 as a percentage of total molecules. The partial pressures of the third characteristic 30 are indicated in FIG. 4 as a percentage of a total pressure.

[0128] A minimum value of the performance-based variable forms part of the third characteristic curve and/or the third characteristic 30. All other values of the performance-based variable along the third characteristic curve and/or along the third characteristic 30 are greater than that minimum value of the performance-based variable.

[0129] The minimum value of the performance-based variable of the third characteristic curve and/or the third characteristic 30 is generally not identical to the corresponding minimum value along the horizontal axis. The minimum value of the performance-based variable of the third char-

acteristic curve and/or the third characteristic 30 can lie, for example, between ten and thirty percent of the nominal performance of the combustion apparatus 1. The nominal performance of the combustion apparatus 1 relates to a nominal heating performance of the combustion apparatus 1.

[0130] In particular, the minimum value of the performance-based variable of the third characteristic curve and/or the third characteristic 30 can lie at twenty percent of the nominal performance of the combustion apparatus 1. Preferably, the minimum value of the performance-based variable of the third characteristic curve and/or the third characteristic 30 is identical to the corresponding minimum value of the first characteristic curve and/or the first characteristic 28.

[0131] Preferably, the minimum value of the performance-based variable of the third characteristic curve and/or the third characteristic 30 is identical to the corresponding minimum value of the second characteristic curve and/or the second characteristic 29. In the example in FIG. 4 the minimum value of the performance-based variable of the third characteristic curve and/or the third characteristic 30 lies at twenty percent of the nominal performance.

[0132] A maximum value of the performance-based variable forms part of the third characteristic curve and/or the third characteristic 30. All other values of the performance-based variable along the third characteristic curve and/or along the third characteristic 30 are lower and/or smaller than that maximum value of the performance-based variable.

[0133] The maximum value of the performance-based variable of the third characteristic curve and/or the third characteristic 30 is generally not identical to the corresponding maximum value along the horizontal axis. The maximum value of the performance-based variable of the third characteristic curve and/or the third characteristic 30 can lie, for example, between ninety and one hundred and ten percent of the nominal performance of the combustion apparatus 1. In particular, the maximum value of the performance-based variable of the third characteristic curve and/or the third characteristic 30 can lie at one hundred percent of the nominal performance of the combustion apparatus 1. In some embodiments, the maximum value of the performance-based variable of the third characteristic curve and/or the third characteristic 30 is identical to the corresponding maximum value of the first characteristic curve and/or the first characteristic 28. In some embodiments, the maximum value of the performance-based variable of the third characteristic curve and/or the third characteristic 30 is identical to the corresponding maximum value of the second characteristic curve and/or the second characteristic 29. In the example in FIG. 4 the maximum value of the performance-based variable of the third characteristic curve and/or the third characteristic 30 lies at one hundred percent of the nominal performance.

[0134] (Three) further points are plotted along the third characteristic curve and/or the third characteristic 30 between the minimum value of the performance-based variable and the corresponding maximum value. The further points are points and/or values 27 of the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure with respect to the performance-based variable 23. Similarly, that point, which forms part of the minimum value of the performance-based variable, is a point and/or value 27 of the residual oxygen content and/or

the oxygen concentration and/or the oxygen partial pressure with respect to the performance-based variable 23. Further, that point which forms part of the maximum value of the performance-based variable, is a point and/or value 27 of the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure with respect to the performance-based variable 23.

[0135] It is possible to interpolate between the points of the third characteristic curve and/or of the third characteristic 30. For example, it is possible to linearly interpolate between the points of the third characteristic curve and/or the third characteristic 30. Further, minimum values of the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure can be determined using cubic splines for values 23 of the performance-based variable. In particular, minimum values of the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure can be calculated using cubic splines for values 23 of the performance-based variable.

[0136] The third characteristic curve and/or the third characteristic 30 can be a mathematical relationship, such as a polynomial, moreover. Minimum values of the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure are thus calculated using the mathematical relationship for values 23 of the performance-based variable. In particular, minimum values of the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure are calculated using a polynomial for values 23 of the performance-based variable. The polynomial corresponds to the third characteristic curve and/or the third characteristic 30.

[0137] The third characteristic curve and/or the third characteristic 30 reproduces a minimum characteristic curve and/or a minimum characteristic of the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure. That is to say, the closed-loop control facility and/or open-loop control facility and/or monitoring facility 13 is or can be communicatively connected to a sensor 20, for example to an oxygen sensor 20, of the combustion apparatus 1. In addition, the closed-loop control facility and/or open-loop control facility and/or monitoring facility 13 is or can be communicatively connected to at least one fuel actuator 7-9 of the combustion apparatus 1. A third characteristic curve and/or a third characteristic 30 is stored in a memory, for example in a non-volatile memory, of the closed-loop control facility and/or open-loop control facility and/or monitoring facility 13. In some embodiments, the closed-loop control facility and/or open-loop control facility and/or monitoring facility 13 is embodied:

[0138] to receive one or more signal(s), which indicate(s) a residual oxygen content and/or an oxygen concentration and/or an oxygen partial pressure, from the sensor 20;

[0139] to determine a measured value of the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure on the basis of the one or more signal(s), which indicate(s) the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure;

[0140] to load the third characteristic curve and/or the third characteristic 30 from the memory;

[0141] to map a burner performance 23, in particular a current burner performance 23, to a comparative value 27 of the residual oxygen content and/or the oxygen

concentration and/or of the oxygen partial pressure using the third characteristic curve and/or using the third characteristic 30;

[0142] to compare the comparative value 27 with the measured value; and

[0143] if the measured value is lower and/or smaller than the comparative value 27;

[0144] to generate a closure signal and send it to the at least one fuel actuator 7-9, wherein the closure signal causes the at least one fuel actuator 7-9 to close when the at least one fuel actuator 7-9 receives the signal.

[0145] The closed-loop control and/or open-loop control illustrated in FIG. 2 is in principle also possible without a sensor 20. That is to say, the closed-loop control and/or open-loop control in FIG. 2 is in principle also possible without a signal which indicates a residual oxygen content and/or an oxygen concentration and/or an oxygen partial pressure.

[0146] The statements relating to minimum and maximum values along the horizontal axis from FIG. 4 can be applied accordingly to FIG. 5 and FIG. 6. In contrast to FIG. 2, FIG. 5 and FIG. 6 show reduced characteristic curves 35 and 36 for the air actuators 3, 4. This applies to combustion apparatuses 1 with combustion in the presence of a flame. With combustion apparatuses 1 for combustion that avoids emissions of nitrogen oxides, the characteristic curves 35 and 36 would be increased with respect to the corresponding characteristic curves in FIG. 2.

[0147] At least one signal from the sensor 20 is accordingly recorded during operation of the combustion apparatus 1. The at least one signal from the sensor 20 indicates a residual oxygen content and/or an oxygen concentration and/or an oxygen partial pressure.

[0148] The at least one signal from the sensor 20 is sent to the closed-loop control facility and/or open-loop control facility and/or monitoring facility 13. The closed-loop control facility and/or open-loop control facility and/or monitoring facility 13 receives the at least one signal from the sensor 20. The closed-loop control facility and/or open-loop control facility and/or monitoring facility 13 ascertains a measured value from the at least one signal. In some embodiments, this is a measured value of a residual oxygen content and/or an oxygen concentration and/or an oxygen partial pressure.

[0149] In addition, the closed-loop control facility and/or open-loop control facility and/or monitoring facility 13 ascertains a burner performance 23. In some embodiments, the burner performance 23 is a heating performance of the combustion apparatus 1. For example, the burner performance 23 can be ascertained from a request signal. That is to say, a burner performance 23 is requested from the combustion apparatus 1 and a corresponding request signal is sent to the closed-loop control facility and/or open-loop control facility and/or monitoring facility 13.

[0150] In particular, the closed-loop control facility and/or open-loop control facility and/or monitoring facility 13 can ascertain a current burner performance 23 of the combustion apparatus 1. In some embodiments, the current burner performance 23 is a current heating performance of the combustion apparatus 1. For example, the current burner performance 23 can be ascertained from a request signal. That is to say, a current burner performance 23 is requested from the combustion apparatus 1 and a corresponding

request signal sent to the closed-loop control facility and/or open-loop control facility and/or monitoring facility 13.

[0151] A relative fluid performance 31 for the current burner performance 23 indicates by what amount the working point of at least one actuator of the combustion apparatus 1 has to be displaced. In some embodiments, the at least one actuator 3, 4, 7-9 of the combustion apparatus 1 can be selected from:

[0152] a fuel actuator 7-9 of the combustion apparatus 1, wherein the fuel actuator 7-9 acts on a fuel supply to the combustion apparatus 1,

[0153] a fan 3 of the combustion apparatus 1, wherein the fan 3 acts on an air supply to the combustion apparatus 1,

[0154] an air damper 4 of the combustion apparatus 1, wherein the air damper 4 acts on an air supply to the combustion apparatus 1.

[0155] The preceding list of actuators 3, 4, 7-9 is not intended to be exhaustive.

[0156] The current burner performance 23 in the present, open-loop control mode will be referred to hereinafter as the first current value of the performance-based variable 23.

[0157] If the at least one actuator is a fuel actuator 7-9, the relative fluid performance characteristic curve 32 is a relative fuel performance characteristic curve. If the at least one actuator is a fan 3, the relative fluid performance characteristic curve 32 is a relative air performance characteristic curve. If the at least one actuator is an air damper 4, the relative fluid performance characteristic curve 32 is likewise a relative air performance characteristic curve.

[0158] With combustion apparatuses 1 for combustion that avoids emissions of nitrogen oxides the fluid performance characteristic curve 32 would be in the negative region of FIG. 5.

[0159] The statements from FIG. 4 relating to interpolation can accordingly be applied to change characteristic curve 32 and/or the relative fluid performance characteristic curve 32 from FIG. 5. The statements from FIG. 4 relating to curves 24-26 in the form of polynomials can accordingly be applied to the change characteristic curve 32 and/or the relative fluid performance characteristic curve 32 from FIG. 5. That is to say, the change characteristic curve 32 and/or the relative fluid performance characteristic curve 32 can be stored as a polynomial in the closed-loop control facility and/or open-loop control facility and/or monitoring facility 13. They can be stored in a memory, such as a non-volatile memory.

[0160] The displacement of the working point means that a new working point of the at least one actuator 3, 4, 7-9 is ascertained for a burner performance 23 using the relative fluid performance 31. The new working point of the at least one actuator 3, 4, 7-9 indicates a performance which is different from the burner performance 23.

[0161] The displacement of the working point can also mean that a new working point of the at least one actuator 3, 4, 7-9 is ascertained for a current burner performance 23 using the relative fluid performance 31. The new working point of the at least one actuator 3, 4, 7-9 indicates a performance which is different from the current burner performance 23.

[0162] If the at least one actuator is a fuel actuator 7-9, the working point of the at least one actuator 7-9 is displaced in the direction of a lower performance.

[0163] The working point of the fuel actuator 7-9 can be displaced within five seconds of the start-up of the combustion apparatus 1. The working point of the fuel actuator 7-9 can be displaced within a second or within two seconds of the start-up of the combustion apparatus 1. The working point of the fuel actuator 7-9 can be displaced instantaneously at the start-up of the combustion apparatus 1. That is to say, the working point of the fuel actuator 7-9 is displaced shortly after or at the start-up of combustion in the combustion apparatus 1.

[0164] The prompt displacement of the working point avoids combustion with harmful emissions when the combustion apparatus 1 is started. The prompt displacement of the working point enables a safe start-up of the combustion apparatus 1 with acceptable emissions.

[0165] FIG. 5 illustrates that displacement of the working point of the fuel actuator 7-9 using the arrow 33. For a burner performance 23, the working point of the fuel actuator 7-9 is thus displaced in the direction of a lower performance using the arrow 33. The displacement is based on the relative fluid performance characteristic curve 32. Instead of a characteristic curve 32, the working point can also be displaced by a constant.

[0166] In particular, for a current burner performance 23 the working point of the fuel actuator 7-9 can be displaced using the arrow 33 in the direction of a lower performance. The displacement is based on the relative fluid performance characteristic curve 32.

[0167] If the at least one actuator is a fan 3 and/or an air damper 4, the working point of the at least one actuator 7-9 is displaced in the direction of a higher performance.

[0168] The working point of the fan 3 can be displaced within five seconds of the start-up of the combustion apparatus 1. The working point of the fan 3 can be displaced within one second or within two seconds of the start-up of the combustion apparatus 1. The working point of the fan 3 can be displaced instantaneously at the start-up of the combustion apparatus 1. That is to say, the working point of the fan 3 is displaced shortly after or at the start-up of combustion in the combustion apparatus 1.

[0169] The working point of the air damper 4 can be displaced within five seconds of the start-up of the combustion apparatus 1. The working point of the air damper 4 can be displaced within one second or within two seconds of the start-up of the combustion apparatus 1. The working point of the air damper 4 can be displaced instantaneously at the start-up of the combustion apparatus 1. That is to say, the working point of the air damper 4 can be displaced shortly after or at the start-up of combustion of the combustion apparatus 1.

[0170] The prompt displacement of the working point avoids combustion with harmful emissions when the combustion apparatus 1 is started. The prompt displacement of the working point enables a safe start-up of the combustion apparatus 1 with acceptable emissions.

[0171] FIG. 5 illustrates that displacement of the working point of the fan 3 or the air damper 4 using the arrow 34. The working point of the fan 3 and/or the air damper 4 is thus displaced using the arrow 34 in the direction of a higher performance for a burner performance 23. The displacement is based on the relative fluid performance characteristic curve 32. Instead of a characteristic curve 32, the working point can also be displaced by a constant.

[0172] In particular, for a current burner performance 23 the working point of the fan 3 and/or the air damper 4 can be displaced using the arrow 34 in the direction of a higher performance. The displacement is based on the relative fluid performance characteristic curve 32.

[0173] That is to say, the closed-loop control facility and/or open-loop control facility and/or monitoring facility 13 is or can be communicatively connected to a sensor 20, for example to an oxygen sensor 20, of the combustion apparatus 1. In addition, the closed-loop control facility and/or open-loop control facility and/or monitoring facility 13 is or can be communicatively connected to at least one actuator 3, 4, 7-9 of the combustion apparatus 1. In some embodiments, the at least one actuator 3, 4, 7-9 of the combustion apparatus 1 is selected from:

[0174] a fuel actuator 7-9 of the combustion apparatus 1, wherein the fuel actuator 7-9 acts on a fuel supply to the combustion apparatus 1,

[0175] a fan 3 of the combustion apparatus 1, wherein the fan 3 acts on an air supply to the combustion apparatus 1,

[0176] an air damper 4 of the combustion apparatus 1, wherein the air damper 4 acts on an air supply to the combustion apparatus 1.

[0177] In some embodiments, there is a closed-loop control and/or open-loop control and/or monitoring facilities 13 with the inclusion of a displaced working point, wherein the closed-loop control facilities and/or open-loop control facilities and/or monitoring facilities 13 is embodied:

[0178] to calculate the displaced working point as the sum of the burner performance 23 and the relative fluid performance, in particular as the sum of the current burner performance 23 and the relative fluid performance; and

[0179] to generate as a function of the displaced working point an open-loop control signal and/or closed-loop control signal and to send the open-loop control and/or closed-loop control signal to the at least one actuator 3, 4, 7-9.

[0180] In some embodiments, there is a closed-loop control and/or open-loop control and/or monitoring facilities 13 with the inclusion of a displaced working point, wherein the closed-loop control facilities and/or open-loop control facilities and/or monitoring facilities 13 is embodied:

[0181] to calculate the displaced working point by scaling the burner performance 23 using a positive scale factor, in particular by scaling the current burner performance 23 using a positive scale factor; and

[0182] to generate as a function of the displaced working point an open-loop control signal and/or closed-loop control signal and to send the open-loop control and/or closed-loop control signal to the at least one actuator 3, 4, 7-9.

[0183] In some embodiments, the scaling comprises a multiplication. In some embodiments, the scaling is a multiplication. In the present case the scale factor is greater than one.

[0184] In some embodiments, the closed-loop control and/or open-loop control and/or monitoring facilities 13 with the inclusion of a displaced working point, wherein the closed-loop control facilities and/or open-loop control facilities and/or monitoring facilities 13 is embodied:

[0185] to calculate the displaced working point as the difference between the burner performance 23 and the

relative fluid performance, in particular as the difference between current burner performance 23 and the relative fluid performance; and

[0186] to generate as a function of the displaced working point an open-loop control signal and/or closed-loop control signal and to send the open-loop control and/or closed-loop control signal to the at least one actuator 3, 4, 7-9.

[0187] In some embodiments, there is a closed-loop control and/or open-loop control and/or monitoring facilities 13 with the inclusion of a displaced working point, wherein the closed-loop control facilities and/or open-loop control facilities and/or monitoring facilities 13 is embodied:

[0188] to calculate the displaced working point by scaling the burner performance 23 using a positive scale factor, in particular by scaling the burner performance 23 using a positive scale factor; and

[0189] to generate as a function of the displaced working point an open-loop control signal and/or closed-loop control signal and to send the open-loop control and/or closed-loop control signal to the at least one actuator 3, 4, 7-9.

[0190] In some embodiments, the scaling comprises a multiplication. In some embodiments, the scaling is a multiplication. In the present case the scale factor is less than one.

[0191] Using the characteristic curves 24-26 illustrated in FIG. 2 combustion may be subject to open-loop control at the start-up of the combustion apparatus 1 such that an adequate excess of air exists. That is to say, the characteristic curves 24 and 26 for air and the characteristic curve 25 for fuel imply an excess of air.

[0192] Such open-loop control avoids combustion with harmful emissions at the start-up of the combustion apparatus 1. Such open-loop control likewise enables a safe start-up of the combustion apparatus 1 with acceptable emissions. Such open-loop control also avoids combustion with harmful emissions in an emergency mode of the combustion apparatus 1. Such open-loop control likewise enables an emergency mode of the combustion apparatus 1 with acceptable emissions. Such an emergency mode can be caused, for example, by failure of an O₂ closed-loop control.

[0193] In contrast to FIG. 2, FIG. 5 and FIG. 6 show reduced characteristic curves 35 and 36 for the air actuators 3, 4.

[0194] In contrast to FIG. 2, FIG. 6 illustrates closed-loop control and/or open-loop control using a signal of the sensor 20. That is to say, the closed-loop control and/or open-loop control in FIG. 6 includes a signal, which indicates a residual oxygen content and/or an oxygen concentration and/or an oxygen partial pressure.

[0195] Specifically, FIG. 5 and FIG. 6 show a characteristic curve 35 for the motor-driven fan 3. Characteristic curve 35 consequently illustrates a speed of the motor-driven fan 3 against the burner performance 23. In particular, characteristic curve 35 illustrates a speed of the motor-driven fan 3 against a current burner performance 23.

[0196] The characteristic curve 35 in FIG. 5 and in FIG. 6 is shifted downwards with respect to the characteristic curve 24 in FIG. 2. That is to say, the characteristic curve 35 in FIG. 5 and in FIG. 6 is displaced to lower values of the positions and/or speeds 22 with respect to the characteristic curve 24 in FIG. 2.

[0197] The statements from FIG. 4 relating to interpolation can be applied accordingly to the characteristic curve 35 for the motor-driven fan 3 from FIG. 5 and from FIG. 6. The statements from FIG. 4 relating to curves 24-26 in the form of polynomials can be applied accordingly to the characteristic curve 35 for the motor-driven fan 3 from FIG. 5 and FIG. 6. That is to say, the characteristic curve 35 for the motor-driven fan 3 can be stored as a polynomial in the closed-loop control facility and/or open-loop control facility and/or monitoring facility 13. The characteristic curve can be stored in a memory, such as a non-volatile memory.

[0198] FIG. 5 and FIG. 6 also show a characteristic curve 36 for an air damper 4. Characteristic curve 36 consequently illustrates a position of the air damper 4 against the burner performance 23. In particular, characteristic curve 36 illustrates a position of the air damper 4 against a current burner performance 23.

[0199] The characteristic curve 36 in FIG. 5 and in FIG. 6 is shifted downwards with respect to the characteristic curve 26 in FIG. 2. That is to say, the characteristic curve 36 in FIG. 5 and in FIG. 6 is displaced in the direction of lower values of the positions 22 with respect to the characteristic curve 26 in FIG. 2.

[0200] The statements from FIG. 4 relating to interpolation can be applied accordingly to the characteristic curve 36 for the air damper 4 from FIG. 5 and from FIG. 6. The statements from FIG. 4 relating to curves 24-26 in the form of polynomials can be applied accordingly to the characteristic curve 36 for the air damper 4 from FIG. 5 and from FIG. 6. That is to say, the characteristic curve 36 for the air damper 4 can be stored as a polynomial in the closed-loop control facility and/or open-loop control facility and/or monitoring facility 13. The characteristic curve can be stored in a memory, such as a non-volatile memory.

[0201] Using the characteristic curves 25, 35, 36 from FIG. 5 and FIG. 6, combustion may not always be subject to open-loop control at the start of the combustion apparatus 1 such that an adequate excess of air exists. That is to say, the characteristic curves 35 and 36 for air and the characteristic curve 25 for fuel do not imply an adequate excess of air under all ambient conditions. Open-loop control using solely the characteristic curves 25, 35 and 36 does not enable a safe start-up of the combustion apparatus 1 with acceptable emissions under all ambient conditions. Such an open-loop control process does not enable an emergency mode of the combustion apparatus 1 with acceptable emissions under all ambient conditions. Such an emergency mode can be caused, for example, by failure of an O₂ closed-loop control.

[0202] It is therefore possible to work with an additional excess of air for open-loop control mode at the start-up of the combustion apparatus 1 or in emergency mode. That is to say, with a given burner performance 23 the air actuators 3, 4 can be subject to open-loop control for a slightly higher value of the air supply $\dot{V}_L$. In particular, with a current burner performance 23 the air actuators 3, 4 can be subject to open-loop control for a slightly higher value of the air supply $\dot{V}_L$.

[0203] The statements relating to minimum and maximum values along the horizontal axis from FIG. 4 can be applied accordingly to FIG. 5 and to FIG. 6.

[0204] Specifically, the air actuators 3, 4 for a burner performance P can be subject to open-loop control for a slightly higher burner performance

$$P_{3,4} = P + \Delta P$$

[0205] The additional performance ΔP can lie, for example, between five and thirty percent of the nominal performance of the combustion apparatus **1**. In some embodiments, the additional performance ΔP can lie between ten and twenty percent of the nominal performance of the combustion apparatus **1**. This open-loop control for the higher burner performance may occur independently of the signal of the sensor **20**. In some embodiments, the additional performance ΔP is a function of the burner performance **23**. In some embodiments, the additional performance ΔP is constant.

[0206] In some embodiments, the air actuators **3, 4** for a current burner performance P can be subject to open-loop control for a slightly higher current burner performance

$$P_{3,4} = P + \Delta P$$

[0207] The additional performance ΔP can lie, for example, between five and thirty percent. In some embodiments, the additional performance ΔP can lie between ten and twenty percent. This open-loop control for the higher current burner performance ideally occurs independently of the signal of the sensor **20**. According to one embodiment, the additional performance ΔP is a function of the current burner performance **23**. In some embodiments, the additional performance ΔP is constant.

[0208] Further, the air actuators **3, 4** for a burner performance P can be subject to open-loop control for a slightly higher burner performance

$$P_{3,4} = P \cdot (1 + \text{relLuft})$$

[0209] The relative air performance relLuft as part of a scale factor $(1 + \text{relLuft})$ can lie, for example, between five and thirty percent. In some embodiments, the relative air performance relLuft can lie between ten and twenty percent. In some embodiments, the open-loop control for the higher burner performance occurs independently of the signal of the sensor **20**.

[0210] In some embodiments, the relative air performance relLuft is a function of the burner performance **23**. In some embodiments, the relative air performance relLuft is constant.

[0211] In particular, the air actuators **3, 4** for a current burner performance P can be subject to open-loop control for a slightly higher burner performance

$$P_{3,4} = P \cdot (1 + \text{relLuft})$$

[0212] The relative air performance relLuft as part of a scale factor $(1 + \text{relLuft})$ can lie, for example, between five and thirty percent. In some embodiments, the relative air performance relLuft can lie between ten and twenty percent. In some embodiments, the open-loop control for the higher burner performance occurs independently of the signal of

the sensor **20**. In some embodiments, the relative air performance relLuft is a function of the current burner performance **23**. In some embodiments, the relative air performance relLuft is constant.

[0213] Further, the air actuators **3, 4** can be subject to open-loop control such that an additional air supply $\dot{V}_{L,L}$ is added to an air supply $\dot{V}_L$ for a burner performance **23**:

$$\dot{V}_{L3,4} = \dot{V}_L + \Delta \dot{V}_L$$

[0214] The additional air supply $\dot{V}_{L,L}$ can lie, for example, between five and thirty percent of a nominal value of the air supply $\dot{V}_L$. In some embodiments, the additional air supply $\dot{V}_{L,L}$ can lie between ten and twenty percent of the nominal value of the air supply $\dot{V}_L$. In some embodiments, the nominal value of the air supply $\dot{V}_L$ corresponds to the nominal performance of the combustion apparatus **1**. In some embodiments, the open-loop control for the additional air supply occurs independently of the signal of the sensor

20. In some embodiments, the additional air supply $\dot{V}_{L,L}$ is a function of the burner performance **23**. According to another embodiment, the additional air supply $\dot{V}_{L,L}$ is constant.

[0215] In particular, the air actuators **3, 4** can be subject to open-loop control such that an additional air supply $\dot{V}_{L,L}$ is added to an air supply $\dot{V}_L$ for a current burner performance **23**:

$$\dot{V}_{L3,4} = \dot{V}_L + \Delta \dot{V}_L$$

[0216] The additional air supply $\dot{V}_{L,L}$ can lie, for example, between five and thirty percent of a nominal value of the air supply $\dot{V}_L$. In some embodiments, the additional air supply $\dot{V}_{L,L}$ can lie between ten and twenty percent of the nominal value of the air supply $\dot{V}_L$. In some embodiments, the nominal value of the air supply $\dot{V}_L$ corresponds to the nominal performance of the combustion apparatus **1**. In some embodiments, the open-loop control for the additional air supply occurs independently of the signal of the sensor **20**. According to one embodiment, the additional air supply $\dot{V}_{L,L}$ is a function of the current burner performance **23**. In some embodiments, the additional air supply $\dot{V}_{L,L}$ is constant.

[0217] Furthermore, the air actuators **3, 4** can be subject to open-loop control such that an additional air supply is added to an air supply $\dot{V}_L$ for a burner performance **23**:

$$\dot{V}_{L3,4} = \dot{V}_L \cdot (1 + \text{relLuft})$$

[0218] The relative air performance relLuft as part of a scale factor $(1 + \text{relLuft})$ can lie, for example, between five and thirty percent. In some embodiments, the relative air performance relLuft can lie between ten and twenty percent. This open-loop control for the higher burner performance

ideally occurs independently of the signal of the sensor **20**. In some embodiments, the relative air performance $relLuft$ is a function of the burner performance **23**. In some embodiments, the relative air performance $relLuft$ is constant.

[0219] In particular, the air actuators **3, 4** can be subject to open-loop control such that an additional air supply is added to an air supply $\dot{V}_L$ for a current burner performance **23**:

$$\dot{V}_{L3,4} = \dot{V}_L \cdot (1 + relLuft)$$

[0220] The relative air performance $relLuft$ as part of a scale factor $(1+relLuft)$ can lie, for example, between five and thirty percent. In some embodiments, the relative air performance $relLuft$ can lie between ten and twenty percent. In some embodiments, this open-loop control for the higher burner performance occurs independently of the signal of the sensor **20**. In some embodiments, the relative air performance $relLuft$ is a function of the current burner performance **23**. In some embodiments, the relative air performance $relLuft$ is constant.

[0221] Instead of the air actuators **3, 4**, the fuel actuator **7-9** for a burner performance P can also be subject to open-loop control for a slightly lower burner performance

$$P_9 = P - \Delta P$$

[0222] The performance ΔP to be deducted can lie, for example, between five and thirty percent of the nominal performance of the combustion apparatus **1**. In some embodiments, the performance ΔP to be deducted can lie between ten and twenty percent of the nominal performance of the combustion apparatus **1**. In some embodiments, this open-loop control for the lower burner performance occurs independently of the signal of the sensor **20**. In some embodiments, the performance ΔP to be deducted is a function of the burner performance **23**. In some embodiments, the performance ΔP to be deducted is constant.

[0223] In particular, instead of the air actuators **3, 4**, the fuel actuator **7-9** for a current burner performance P can also be subject to open-loop control for a slightly lower current burner performance

$$P_9 = P - \Delta P$$

[0224] The performance ΔP to be deducted can lie, for example, between five and thirty percent of the nominal performance of the combustion apparatus **1**. In some embodiments, the performance ΔP to be deducted can lie between ten and twenty percent of the nominal performance of the combustion apparatus **1**. In some embodiments, this open-loop control for the lower current burner performance occurs independently of the signal of the sensor **20**. In some embodiments, the performance ΔP to be deducted is a function of the current burner performance **23**. In some embodiments, the performance ΔP to be deducted is constant.

[0225] Further, instead of the air actuators **3, 4**, the fuel actuator **7-9** for a burner performance P can also be subject to open-loop control for a slightly lower burner performance

$$P_9 = P \cdot (1 - relLuft)$$

[0226] The relative air performance $relLuft$ as part of a scale factor $(1-relLuft)$ can lie, for example, between five and thirty percent. In some embodiments, the relative air performance $relLuft$ can lie between ten and twenty percent. In some embodiments, this open-loop control for the lower burner performance occurs independently of the signal of the sensor **20**. In some embodiments, the relative air performance $relLuft$ is a function of the burner performance **23**. In some embodiments, the relative air performance $relLuft$ is constant.

[0227] In particular, instead of the air actuators **3, 4**, the fuel actuator **7-9** for a current burner performance P can also be subject to open-loop control for a slightly lower current burner performance

$$P_9 = P \cdot (1 - relLuft)$$

[0228] The relative air performance $relLuft$ as part of a scale factor $(1-relLuft)$ can lie, for example, between five and thirty percent. In some embodiments, the relative air performance $relLuft$ can lie between ten and twenty percent. In some embodiments, this open-loop control for the lower current burner performance occurs independently of the signal of the sensor **20**. In some embodiments, the relative air performance $relLuft$ is a function of the current burner performance **23**. In some embodiments, the relative air performance $relLuft$ is constant.

[0229] Further, the fuel actuator **7-9** can be subject to open-loop control such that a fuel supply $\dot{V}_B$ is deducted from a fuel supply $\dot{V}_B$ for a burner performance **23**:

$$\dot{V}_{B9} = \dot{V}_B - \Delta \dot{V}_B$$

[0230] The fuel supply $\dot{V}_{L,B}$ to be deducted can lie, for example, between five and thirty percent of a nominal value of the fuel supply $\dot{V}_B$. In some embodiments, the fuel supply $\dot{V}_{L,B}$ to be deducted can lie between ten and twenty percent of the nominal value of the fuel supply $\dot{V}_B$. In some embodiments, the nominal value of the fuel supply $\dot{V}_B$ corresponds to the nominal performance of the combustion apparatus **1**. In some embodiments, this open-loop control for the lower fuel supply $\dot{V}_B$ occurs independently of the signal of the sensor **20**. In some embodiments, the fuel supply $\dot{V}_{L,B}$ to be deducted is a function of the burner performance **23**. In some embodiments, the fuel supply $\dot{V}_{L,B}$ to be deducted is constant.

[0231] In particular, the fuel actuator **7-9** can be subject to open-loop control such that a fuel supply $\dot{V}_B$ is deducted from a fuel supply $\dot{V}_B$ for a current burner performance **23**:

$$\dot{V}_{B9} = \dot{V}_B - \Delta \dot{V}_B$$

[0232] The fuel supply $\dot{V}_{LB}$ to be deducted can lie, for example, between five and thirty percent of a nominal value of the fuel supply $\dot{V}_B$. In some embodiments, the fuel supply $\dot{V}_{LB}$ to be deducted can lie between ten and twenty percent of the nominal value of the fuel supply $\dot{V}_B$. In some embodiments, the nominal value of the fuel supply $\dot{V}_B$ corresponds to the nominal performance of the combustion apparatus 1. In some embodiments, this open-loop control for the lower fuel supply $\dot{V}_B$ occurs independently of the signal of the sensor 20. In some embodiments, the fuel supply $\dot{V}_{LB}$ to be deducted is a function of the current burner performance 23. In some embodiments, the fuel supply $\dot{V}_{LB}$ to be deducted is constant.

[0233] The open-loop control mode at the start-up of the combustion apparatus 1 is followed by a closed-loop control mode of the combustion apparatus 1. In some embodiments, open-loop control mode occurs independently of a signal of the sensor 20. By contrast, closed-loop control can occur, for example, using the sensor 20 of FIG. 3. The transition between subject to open-loop control and closed-loop control mode can occur, for example, at least ten seconds or at least twenty seconds or at least thirty seconds after the start-up of the combustion apparatus 1. In general the instant of the transition between open-loop and closed-loop control mode depends on the dead times of the closed-loop control of the combustion apparatus 1. That is to say, closed-loop control mode of the combustion apparatus 1 occurs after the start of combustion in the combustion apparatus 1.

[0234] For the burner performance 23 the closed-loop control facility and/or open-loop control facility and/or monitoring facility 13 ascertains, as shown in FIG. 4, a desired value 27. The desired value indicates a residual oxygen content and/or an oxygen concentration and/or an oxygen partial pressure. In one embodiment the burner performance 23 is mapped to the desired value 27 using the second characteristic curve and/or using the second characteristic 29 of FIG. 4.

[0235] In particular, the closed-loop control facility and/or open-loop control facility and/or monitoring facility 13 can ascertain a desired value 27, as shown in FIG. 4, for the current burner performance 23. The desired value 27 indicates a residual oxygen content and/or an oxygen concentration and/or an oxygen partial pressure. In some embodiments, the current burner performance 23 is mapped to the desired value 27 using the second characteristic curve and/or using the second characteristic 29 of FIG. 4.

[0236] The current burner performance 23 in closed-loop control mode will hereinafter be referred to as the second, current value of the performance-based variable 23.

[0237] A first actuator, such as an air actuator 3, 4, is subject to closed-loop control. A second actuator, such as the fuel actuator 7-9, is subject to open-loop control during closed-loop control of the first actuator. The second actuator is then subject to open-loop control using one of the characteristic curves 35, 36, 25 of FIG. 5 and FIG. 6.

[0238] In addition, closed loop control mode can occur when a decrease or an increase is requested, for example of the burner performance 23 or the current burner perfor-

mance 23, temporarily with the inclusion of relative fluid performance. In such a case the relative fluid performance is usually a function of the burner performance 23 or the current burner performance 23. The combustion apparatus 1 is operated in closed loop control mode with the inclusion of a relative fluid performance until the closed-loop control after the decrease or increase is sufficiently stable again.

[0239] In closed loop control mode the displaced characteristic curves 35 and 36 may prove to be advantageous for the air actuators 3, 4. These characteristic curves 35, 36 namely correspond to a lower excess of air, as is typically adjusted in closed loop control mode. That is to say, closed-loop control, in contrast to the open-loop control of FIG. 2, does not have to alter the working points of the actuators 3, 4, 7-9, or only has to alter them slightly.

[0240] The above statements relating to an additional performance or a performance ΔP to be deducted can refer to an emergency mode of the combustion apparatus 1. Further, the above statements relating to scaled performances can refer to an emergency mode of the combustion apparatus 1. Similarly, said statements relating to an additional air supply $\dot{V}_{L,L}$ or a fuel supply $\dot{V}_{L,B}$ to be deducted can refer to the emergency mode of the combustion apparatus 1. Further, said statements relating to scaled air or fuel supplying can refer to an emergency mode of the combustion apparatus 1. Combustion with harmful emissions in the emergency mode of the combustion apparatus 1 is thus avoided. A safe emergency mode of the combustion apparatus 1 with acceptable emissions is enabled. Such an emergency mode can be caused, for example, by a failure of an O_2 closed-loop control.

[0241] The current burner performance 23 in emergency mode will hereinafter be referred to as the third, current value of the performance-based variable 23.

[0242] In other words, some embodiments of the teachings of the present disclosure include a method for control of a combustion apparatus (1), the combustion apparatus (1) comprising a combustion chamber (2), an air supply duct (11) leading to the combustion chamber (2), a fuel supply duct leading to the combustion chamber (2), at least one first actuator (3, 4, 7-9) selected from at least one air actuator (3, 4), which acts on an air supply $\dot{V}_L$ through the air supply duct (11), and at least one fuel actuator (7-9), which acts on a fuel supply $\dot{V}_B$ through the fuel supply duct, the combustion apparatus (1) comprising a waste gas path (10), at least one oxygen-based sensor (20) in the waste gas path (10) and a closed-loop control facility and/or open-loop control facility and/or monitoring facility (13) with a memory in which at least one first characteristic curve (25, 35, 36), which for the at least one first actuator (3, 4, 7-9) indicates a first speed characteristic and/or a first position characteristic with respect to a performance-based variable (23), and a change, which is different from zero, are stored, the method comprising the following steps:

[0243] loading the at least one first characteristic curve (25, 35, 36) and the change from the memory;

[0244] determining a first, current value of the performance-based variable (23);

[0245] determining a first input value of the performance-based variable (23) for the open-loop control mode as a function of the first, current value of the performance-based variable (23) and as a function of

the change, wherein the determination occurs independently of an oxygen-based signal of the at least one sensor (20);

[0246] assigning the first input value of the performance-based variable (23) for the open-loop control mode to a first speed and/or to a first position using the at least one first characteristic curve (25, 35, 36);

[0247] determining a first open-loop control signal as a function of the first speed and/or the first position; and

[0248] sending the first open-loop control signal to the at least one first actuator (3, 4, 7-9), wherein the first open-loop control signal causes the at least one first actuator (3, 4, 7-9) to change at least one combustion variable selected from the air supply $\dot{V}_L$ and/or the fuel supply $\dot{V}_B$.

[0249] The at least one first actuator (3, 4, 7-9) and the at least one sensor (20) and the memory are each communicatively connected to the closed-loop control facility and/or open-loop control facility and/or monitoring facility (13).

[0250] In some embodiments, the performance-related variable (23) is selected from

[0251] a burner performance (23) of the combustion apparatus (1);

[0252] a current burner performance (23) of the combustion apparatus (1).

[0253] The at least one sensor (20) is embodied for recording at least one oxygen-based signal and for sending the at least one oxygen-based signal to the closed-loop control facility and/or open-loop control facility and/or monitoring facility (13). The oxygen-based signal can indicate, for example,

[0254] a residual oxygen content and/or

[0255] an oxygen concentration and/or

[0256] an oxygen partial pressure.

[0257] The determination of the first input value independently of an oxygen-based signal of the at least one sensor (20) is a determination excluding an oxygen-based signal of the at least one sensor (20). The determination of the first input value independently of an oxygen-based signal of the at least one sensor (20) can also be a determination that dispenses with an oxygen-based signal of the at least one sensor (20). The oxygen-based signal of the at least one sensor (20) is advantageously an oxygen-based signal from the at least one sensor (20).

[0258] The input value of the performance-based variable (23) for the open-loop control mode is transferred to the open-loop control as an input. The open-loop control ascertains a speed and/or a position from the input value. The input value is consequently a working point for the open-loop control.

[0259] In some embodiments, the determination of the first input value of the performance-based variable (23) for the open-loop control mode as a function of the first, current value of the performance-based variable (23) and as a function of the change occurs independently of the at least one sensor (20).

[0260] In some embodiments, the method comprises sending the first open-loop control signal to the at least one first actuator (3, 4, 7-9), wherein the first open-loop control signal causes the at least one first actuator (3, 4, 7-9) to change at least one combustion variable selected from the air supply $\dot{V}_L$ or the fuel supply $\dot{V}_B$.

[0261] In some embodiments, the method comprises:

[0262] receiving the first open-loop control signal by way of the at least one first actuator (3, 4, 7-9); and

[0263] in response to receiving the first open-loop control signal by way of the at least one first actuator (3, 4, 7-9), changing at least one combustion variable selected from the air supply $\dot{V}_L$ and/or the fuel supply $\dot{V}_B$ by way of the at least one first actuator (3, 4, 7-9).

[0264] In some embodiments, the change is a first, performance-related change. In some embodiments, the change has the unit of a performance. In some embodiments, the performance-related variable (23) and the input value of the performance-based variable (23) for the open-loop control mode also each have the unit of a performance.

[0265] In some embodiments, the method comprises:

[0266] starting combustion in the combustion apparatus (1); and

[0267] sending the first open-loop control signal to the at least one first actuator (3, 4, 7-9) within fifteen seconds of the start of combustion.

[0268] In some embodiments, the method comprises:

[0269] starting combustion in the combustion apparatus (1); and

[0270] sending the first open-loop control signal to the at least one first actuator (3, 4, 7-9) within ten seconds of the start of combustion.

[0271] In some embodiments, the method comprises:

[0272] starting combustion in the combustion apparatus (1); and

[0273] sending the first open-loop control signal to the at least one first actuator (3, 4, 7-9) within five seconds of the start of combustion.

[0274] The start of combustion of the combustion apparatus (1) can be, in particular, a start of combustion in the combustion chamber (2) of the combustion apparatus (1).

[0275] In some embodiments, the change is different from zero and is constant, and the method comprises:

[0276] determining the first input value of the performance-based variable (23) for the open-loop control mode as a function of the first, current value of the performance-based variable (23) and as a function of the first, constant change, wherein the determination occurs independently of the oxygen-based signal of the at least one sensor (20).

[0277] The determination of the first input value of the performance-based variable (23) for the open-loop control mode as a function of the first, current value of the performance-based variable (23) and as a function of the first, constant change ideally occurs independently of the at least one sensor (20).

[0278] In some embodiments, at least one change characteristic curve (32), which indicates a change characteristic with respect to the performance-based variable (23), is stored in the memory, and the method comprises:

[0279] loading the at least one change characteristic curve (32) from the memory;

[0280] determining the change from the first, current value of the performance-based variable (23) by assigning the first, current value of the performance-based variable (23) using the at least one change characteristic curve (32); and

[0281] determining the first input value of the performance-based variable (23) for the open-loop control mode as a function of the first, current value of the performance-based variable (23) and as a function of

the change from the first, current value of the performance-based variable (23), wherein the determination occurs independently of the oxygen-based signal of the at least one sensor (20).

[0282] The determination of the first input value of the performance-based variable (23) for the open-loop control mode as a function of the first, current value of the performance-based variable (23) and as a function of the change from the first, current value of the performance-based variable (23) ideally occurs independently of the at least one sensor (20).

[0283] In some embodiments, the method comprises determining the first input value of the performance-based variable (23) for the open-loop control mode as the sole function of the first, current value of the performance-based variable (23) and the change. In a sole function there are no function arguments apart from the stated function arguments.

[0284] In some embodiments, the method comprises determining the first input value of the performance-based variable (23) for the open-loop control mode as the sum of the first, current value of the performance-based variable (23) and the change, wherein the determination occurs independently of the oxygen-based signal of the at least one sensor (20).

[0285] In some embodiments, the determination of the first input value of the performance-based variable (23) for the open-loop control mode as the sum of the first, current value of the performance-based variable (23) and the change occurs independently of the at least one sensor (20).

[0286] In some embodiments, the method comprises determining the first input value of the performance-based variable (23) for the open-loop control mode as the difference between the first, current value of the performance-based variable (23) and the change, wherein the determination occurs independently of the oxygen-based signal of the at least one sensor (20). In some embodiments, the determination occurs independently of the at least one sensor (20).

[0287] In some embodiments, at least one desired value characteristic curve (29), which indicates a characteristic of a residual oxygen content and/or an oxygen concentration and/or an oxygen partial pressure with respect to the performance-based variable (23), is stored in the memory, and the method comprises:

[0288] after the first open-loop control signal has been sent, recording at least one oxygen-based signal, which indicates the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure, by way of the at least one sensor (20);

[0289] sending the oxygen-based signal to the closed-loop control facility and/or open-loop control facility and/or monitoring facility (13);

[0290] determining a measured value by way of the closed-loop control facility and/or open-loop control facility and/or monitoring facility (13) on the basis of the at least one oxygen-based signal;

[0291] determining a second, current value of the performance-based variable (23);

[0292] loading the at least one desired value characteristic curve (29) from the memory;

[0293] assigning the second, current value of the performance-based variable (23) to a desired value (27) using the at least one desired value characteristic curve (29);

[0294] comparing the measured value with the desired value (27);

[0295] generating a closed-loop control signal on the basis of the comparison between the measured value and the desired value (27); and

[0296] sending the closed-loop control signal to the at least one first actuator (3, 4, 7-9), wherein the closed-loop control signal causes the at least one first actuator (3, 4, 7-9) to change the at least one combustion variable.

[0297] In some embodiments, the inclusion of at least one desired value characteristic curve (29), the method comprises at least thirty seconds after the first open-loop control signal has been sent, recording at least one oxygen-based signal, which indicates the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure, by way of the at least one sensor (20).

[0298] In some embodiments, the method comprises at least twenty seconds after the first open-loop control signal has been sent, recording at least one oxygen-based signal, which indicates the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure, by way of the at least one sensor (20).

[0299] In some embodiments, the method comprises at least ten seconds after the first open-loop control signal has been sent, recording at least one oxygen-based signal, which indicates the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure, by way of the at least one sensor (20).

[0300] In some embodiments, the method comprises:

[0301] receiving the closed-loop control signal by way of the at least one first actuator (3, 4, 7-9); and

[0302] in response to receiving the closed-loop control signal by way of the at least one first actuator (3, 4, 7-9), changing the at least one combustion variable by way of the at least one first actuator (3, 4, 7-9).

[0303] In some embodiments, the combustion apparatus (1) comprises at least one second actuator (7-9, 3, 4) selected from the at least one fuel actuator (7-9) and the at least one air actuator (3, 4), wherein the at least one second actuator (7-9, 3, 4) is different from the at least one first actuator (3, 4, 7-9), wherein at least one second characteristic curve (35, 36, 25), which for the at least one second actuator (7-9, 3, 4) indicates a second speed characteristic and/or a second position characteristic with respect to the performance-based variable (23), is stored in the memory, and the method comprises:

[0304] assigning the second, current value of the performance-based variable (23) to a second speed and/or to a second position using the at least one second characteristic curve (35, 36, 25), wherein the assignment takes place independently of the change;

[0305] determining a second open-loop control signal as a function of the second speed and/or the second position; and

[0306] sending the second open-loop control signal to the at least one second actuator (7-9, 3, 4), wherein the second open-loop control signal causes the at least one second actuator (7-9, 3, 4) to change the at least one combustion variable.

[0307] The at least one first actuator (3, 4, 7-9) and the at least one second actuator (7-9, 3, 4) and the at least one sensor (20) and the memory are each communicatively connected to the closed-loop control facility and/or open-loop control facility and/or monitoring facility (13).

[0308] The mixed, open-loop and closed-loop control mode enables a prompt reaction of the combustion apparatus (1) to changed ambient conditions. Open-loop control mode occurs independently of the at least one oxygen-based signal of the at least one sensor (20). Closed-loop control mode occurs with the inclusion at least one oxygen-based signal of the at least one sensor (20).

[0309] In some embodiments, the method comprises:

[0310] receiving the second open-loop control signal by way of the at least one second actuator (7-9, 3, 4); and

[0311] in response to receiving the second open-loop control signal by way of the at least one second actuator (7-9, 3, 4), changing the at least one combustion variable by way of the at least one second actuator (7-9, 3, 4).

[0312] In some embodiments, the method comprises determining a second, current value of the performance-based variable (23) from the first, current value of the performance-based variable (23).

[0313] It is possible that the performance requirement of the combustion apparatus (1) does not change with the transition from open-loop to closed-loop control mode. In this case the first, current value of the performance-based variable (23) is identical to the second, current value of the performance-based variable (23).

[0314] The combustion apparatus (1) can consequently optimize the combustion for the initial performance requirement in closed-loop control mode.

[0315] In some embodiments, the method comprises determining a second, current value of the performance-based variable (23) after the first open-loop control signal has been sent.

[0316] In some embodiments, the method comprises determining a second, current value of the performance-based variable (23) at least thirty seconds after the first open-loop control signal has been sent.

[0317] In some embodiments, the method comprises determining a second, current value of the performance-based variable (23) at least twenty seconds after the first open-loop control signal has been sent.

[0318] In some embodiments, the method comprises determining a second, current value of the performance-based variable (23) at least ten seconds after the first open-loop control signal has been sent.

[0319] In some embodiments, the second, current value of the performance-based variable (23) is different from the first, current value of the performance-based variable (23). It is possible that the performance requirement of the combustion apparatus (1) changes with the transition from open-loop to closed-loop control mode. The combustion apparatus (1) can consequently react to changed performance requirements.

[0320] In some embodiments, the method comprises:

[0321] checking the measured value for at least one error;

[0322] if checking of the measured value yields the at least one error;

[0323] determining a third, current value of the performance-based variable (23);

[0324] determining a further input value of the performance-based variable (23) for the open-loop control mode as a function of the second or third current value of the performance-based variable (23) and as a function of the change, wherein the determination occurs independently of the oxygen-based signal of the at least one sensor (20);

[0325] assigning the further input value of the performance-based variable (23) for the open-loop control mode to a third speed and/or to a third position using the at least one first characteristic curve (25, 35, 36);

[0326] determining an emergency open-loop control signal as a function of the third speed and/or the third position; and

[0327] sending the emergency open-loop control signal to the at least one first actuator (3, 4, 7-9), wherein the emergency open-loop control signal causes the at least one first actuator (3, 4, 7-9) to change the at least one combustion variable.

[0328] In some embodiments, a limit value for the measured value is stored in the memory, and the method comprises:

[0329] checking the measured value for the at least one error by comparing the measured value with the limit value; and

[0330] identifying the at least one error if the measured value is smaller than the limit value.

[0331] Incorrect measured values can be, for example, negative residual oxygen contents and/or negative oxygen concentrations and/or negative oxygen partial pressures. In such a case the limit value can be zero.

[0332] In some embodiments, the at least one sensor (20) comprises a digital interface. The at least one sensor (20) can communicate at least one error signal to the closed-loop control facility and/or open-loop control facility and/or monitoring facility (13) via the digital interface. The closed-loop control facility and/or open-loop control facility and/or monitoring facility (13) determines a measured value, which indicates the at least one error, on the basis of the at least one error signal.

[0333] In some embodiments, the method comprises:

[0334] checking the measured value for the at least one error by comparing the measured value with the limit value; and

[0335] identifying the at least one error if the measured value is greater than the limit value.

[0336] For example, a residual oxygen content, which is greater than one hundred percent, points to the at least one error. In addition, a residual oxygen content, which is greater than twenty-one percent, can point to the at least one error.

[0337] Some embodiments include a combustion apparatus (1) comprising a combustion chamber (2), an air supply duct (11) leading to the combustion chamber (2), a fuel supply duct leading to the combustion chamber (2), at least one first actuator (3, 4, 7-9) selected from at least one air actuator (3, 4), which acts on an air supply $\dot{V}_L$ through the air supply duct (11), and at least one fuel actuator (7-9), which acts on a fuel supply $\dot{V}_B$ through the fuel supply duct, the combustion apparatus (1) comprising at least one second actuator (7-9, 3, 4) selected from the at least one fuel actuator (7-9) and the at least one air actuator (3, 4), wherein the at least one second actuator (7-9, 3, 4) is different from the at least one first actuator (3, 4, 7-9), the combustion apparatus (1) comprising a waste gas path (10), at least one

oxygen-based sensor (20) in the waste gas path (10) and a closed-loop control facility and/or open-loop control facility and/or monitoring facility (13) with a memory in which at least one first characteristic curve (25, 35, 36), which indicates for the at least one first actuator (3, 4, 7-9) a first speed characteristic and/or a first position characteristic with respect to a performance-based variable (23), and at least one second characteristic curve (35, 36, 25), which indicates for the at least one second actuator (7-9, 3, 4) a second speed characteristic and/or a second position characteristic with respect to the performance-based variable (23), and a change, which is different from zero, and a limit value for a measured value are stored, wherein the closed-loop control facility and/or open-loop control facility and/or monitoring facility (13) is communicatively connected to the at least one first actuator (3, 4, 7-9) and to the at least one second actuator (7-9, 3, 4) and to the at least one oxygen-based sensor (20) and to the memory and is configured to carry out one of said methods.

[0338] Some embodiments include a computer program comprising instructions which cause the closed-loop control facility and/or open-loop control facility and/or monitoring facility (13) of one of said combustion apparatuses (1) to execute one or more of the methods described herein. Some embodiments include a computer-readable medium on which said computer program or one of said computer programs is stored.

[0339] The aforementioned relates to individual embodiments of the disclosure. Various changes to the embodiments can be made without deviating from the underlying idea and without departing from the context of this disclosure. A wide variety of changes can be made without departing from the scope of the claims.

REFERENCE CHARACTERS

- [0340] 1: combustion apparatus
- [0341] 2: combustion chamber
- [0342] 3: fan with (optionally) changeable speed
- [0343] 4: air damper with actuating drive
- [0344] 5: supply air
- [0345] 6: fuel for combustion
- [0346] 7: safety shut-off valve
- [0347] 8: safety shut-off valve
- [0348] 9: fuel actuator with actuating drive for changing the fuel supply
- [0349] 10: waste gas path, in particular waste gas stack and/or flue gas stack and/or flue
- [0350] 11: air supply duct
- [0351] 12: sensor for detecting the air supply (air mass flow/speed etc.)
- [0352] 13: closed-loop control and/or open-loop control and/or monitoring facility
- [0353] 14: open-loop control signal for air damper (positioning angle)
- [0354] 15: actuating signal for fan speed (optional)
- [0355] 16: measuring signal from the air supply sensor
- [0356] 17: open/close signal for safety shut-off valve
- [0357] 18: open/close signal for safety shut-off valve
- [0358] 19: open-loop control signal for fuel actuator (for example positioning angle/step position)
- [0359] 20: sensor in the waste gas duct
- [0360] 21: measuring signal from the sensor in the waste gas duct
- [0361] 22: positions and/or speeds

- [0362] 23: performance-related variable, in particular burner performance or current burner performance
- [0363] 24: characteristic curve for the motor-driven fan, in particular actuating characteristic curve for the motor-driven fan
- [0364] 25: characteristic curve for the fuel actuator, in particular control characteristic curve for the fuel actuator or closed-loop control characteristic curve for the fuel actuator
- [0365] 26: characteristic curve for the air damper, in particular control characteristic curve for the air damper
- [0366] 27: oxygen concentration and/or oxygen partial pressure and/or residual oxygen content
- [0367] 28: characteristic curve of the maximum oxygen concentration and/or the maximum oxygen partial pressure and/or the maximum residual oxygen content
- [0368] 29: desired value characteristic curve of the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure and/or the residual oxygen content
- [0369] 30: characteristic curve of the minimum oxygen concentration and/or the minimum oxygen partial pressure and/or the minimum residual oxygen content
- [0370] 31: relative fluid performance
- [0371] 32: change characteristic curve, in particular relative fluid performance characteristic curve
- [0372] 33: arrow relating to displacement of the working point of the fuel actuator
- [0373] 34: arrow relating to displacement of the working point of the fan 3 or the air damper 4
- [0374] 35: characteristic curve for the motor-driven fan, in particular closed-loop control characteristic curve for the motor-driven fan
- [0375] 36: characteristic curve for the air damper, in particular closed-loop control characteristic curve for the air damper

1. A method for control of a combustion apparatus comprising a combustion chamber, an air supply duct leading to the combustion chamber, a fuel supply duct leading to the combustion chamber, a first actuator selected from: an air actuator for an air supply through the air supply duct, and a fuel actuator for a fuel supply through the fuel supply duct, a waste gas path, an oxygen-based sensor in the waste gas path, and a control facility with a memory storing a first characteristic curve for the first actuator with a first speed characteristic and/or a first position characteristic with respect to a performance-based variable, and a change, the method comprising:

- loading the first characteristic curve and the change from the memory;
- determining a first, current value of the performance-based variable;
- determining a first input value of the performance-based variable for an open-loop control mode as a function of the first, current value of the performance-based variable and the change, wherein the determination occurs independently of an signal of the oxygen-based sensor;
- assigning the first input value of the performance-based variable to a first speed and/or to a first position using the first characteristic curve;
- determining a first open-loop control signal as a function of the first speed and/or the first position; and

sending the first open-loop control signal to the first actuator, wherein the first open-loop control signal causes the first actuator to change a combustion variable including the air supply and/or the fuel supply.

2. The method as claimed in claim 1, further comprising: starting combustion in the combustion apparatus; and sending the first open-loop control signal to the first actuator within five seconds of the start of combustion.

3. The method as claimed in claim 1, wherein the change is different from zero and is constant;

the method further comprising determining the first input value of the performance-based variable for the open-loop control mode as a function of the first, current value of the performance-based variable and the first, constant change,

wherein the determination occurs independently of the signal of the oxygen-based sensor.

4. The method as claimed in claim 1, wherein the memory stores a change characteristic curve indicating a change characteristic with respect to the performance-based variable;

the method further comprises:

loading the change characteristic curve from the memory; determining the change from the first, current value of the performance-based variable by assigning the first, current value of the performance-based variable using the change characteristic curve; and

determining the first input value of the performance-based variable for the open-loop control mode as a function of the first, current value of the performance-based variable and the change from the first, current value of the performance-based variable;

wherein the determination occurs independently of the signal of the oxygen-based sensor.

5. The method as claimed in claim 1, the method further comprising determining the first input value of the performance-based variable for the open-loop control mode as the sole function of the first, current value of the performance-based variable and the change.

6. The method as claimed in claim 3, the method further comprising determining the first input value of the performance-based variable for the open-loop control mode as the sum of the first, current value of the performance-based variable and the change;

wherein the determination occurs independently of the signal of the oxygen-based sensor.

7. The method as claimed in claim 1, wherein the memory stores at least one desired value characteristic curve indicating a characteristic of a residual oxygen content and/or an oxygen concentration and/or an oxygen partial pressure with respect to the performance-based variable;

the method further comprising:

after the first open-loop control signal has been sent, recording a signal indicating the residual oxygen content and/or the oxygen concentration and/or the oxygen partial pressure, using the oxygen-based sensor;

sending the oxygen-based signal to the control facility; determining a measured value using the control facility on the basis of the oxygen-based signal;

determining a second, current value of the performance-based variable;

loading the desired value characteristic curve from the memory;

assigning the second, current value of the performance-based variable to a desired value using the desired value characteristic curve;

comparing the measured value with the desired value;

generating a closed-loop control signal on the basis of the comparison between the measured value and the desired value; and

sending the closed-loop control signal to the first actuator, wherein the closed-loop control signal causes the first actuator to change the combustion variable.

8. The method as claimed in claim 7, wherein the combustion apparatus comprises a second actuator selected from the fuel actuator and the air actuator different from the first actuator, wherein the memory stores a second characteristic curve for the second actuator indicating a second speed characteristic and/or a second position characteristic with respect to the performance-based variable;

the method further comprising:

assigning the second, current value of the performance-based variable to a second speed and/or to a second position using the second characteristic curve, wherein the assignment takes place independently of the change;

determining a second open-loop control signal as a function of the second speed and/or the second position; and sending the second open-loop control signal to the second actuator, wherein the second open-loop control signal causes the second actuator to change the combustion variable.

9. The method as claimed in claim 7, the method comprising determining a second, current value of the performance-based variable from the first, current value of the performance-based variable.

10. The method as claimed in claim 7, the method comprising determining a second, current value of the performance-based variable after the first open-loop control signal has been sent.

11. The method as claimed in claim 7, the method further comprising:

checking the measured value for an error;

if checking of the measured value yields the error:

determining a third, current value of the performance-based variable;

determining a further input value of the performance-based variable for the open-loop control mode as a function of the second or third current value of the performance-based variable and the change, wherein the determination occurs independently of the signal of the oxygen-based sensor;

assigning the further input value of the performance-based variable to a third speed and/or to a third position using the first characteristic curve;

determining an emergency open-loop control signal as a function of the third speed and/or the third position; and sending the emergency open-loop control signal to the first actuator, wherein the emergency open-loop control signal causes the first actuator to change the combustion variable.

12. The method as claimed in claim 11, wherein the memory stores a limit value for the measured value;

the method further comprising:

checking the measured value for the error by comparing the measured value with the limit value; and

identifying the error if the measured value is smaller than the limit value.

13. A combustion apparatus comprising:

- a combustion chamber;
- an air supply duct leading to the combustion chamber;
- a fuel supply duct leading to the combustion chamber;
- a first actuator selected from an air actuator for an air supply through the air supply duct and a fuel actuator for a fuel supply through the fuel supply duct;
- a second actuator selected from the fuel actuator and the air actuator different from the first actuator;
- a waste gas path;
- an oxygen-based sensor in the waste gas path; and
- a control facility with a memory storing a first characteristic curve indicating for the first actuator a first speed characteristic and/or a first position characteristic with respect to a performance-based variable, and a second characteristic curve indicating for the second actuator a second speed characteristic and/or a second position characteristic with respect to the performance-based variable, a change, and a limit value for a measured value;

wherein the control facility communicates with the first actuator, the second actuator, the oxygen-based sensor, and the memory and is configured to:

- load the first characteristic curve and the change from the memory;
- determine a first, current value of the performance-based variable;
- determine a first input value of the performance-based variable for an open-loop control mode as a function of the first, current value of the performance-based variable and the change, wherein the determination occurs independently of an signal of the oxygen-based sensor;
- assign the first input value of the performance-based variable to a first speed and/or to a first position using the first characteristic curve;
- determine a first open-loop control signal as a function of the first speed and/or the first position; and
- send the first open-loop control signal to the first actuator, wherein the first open-loop control signal causes the first actuator to change a combustion variable including the air supply and/or the fuel supply.

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